Magmatic flow paths and palaeomagnetism of the Miocene Stoddard Mountain laccolith, Iron Axis region, Southwestern Utah, USA

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Abstract: The Stoddard Mountain laccolith is part of a complex of Early Miocene laccoliths intruded along the western edge of the Colorado Plateau in the Iron Axis region of Southwestern Utah. Most Colorado Plateau laccoliths (e.g. Henry and La Sal Mountains) are considered to be fed by a central axial feeder system. However, detailed mapping in the Iron Axis region suggests that the Stoddard Mountain laccolith was fed laterally from the west. Structural and field data suggest the quartz monzonitic magma initially migrated laterally eastward at ~1 km depth as a sill before spreading laterally north-south where it inflated to ~1–1.5 km thickness. To test the model of a lateral feeder system, data were collected from 32 palaeomagnetic sites and 76 AMS stations (763 accepted specimens) sampled over the ~54 km^2 exposed part of the N–S oval-shaped laccolith. The *in situ* AMS fabrics, inferred to correlate with magmatic fabrics, typically show NE trending lineations in the north and S–SE trending lineations in the south part of the intrusion. The palaeomagnetic data are interpreted to indicate a very minimal amount of post-emplacement deformation of the intrusion. The overall lack of westerly-directed and steep magnetic lineations argues against emplacement via a central axial feeder system.

Laccoliths have been well documented and studied for over a century (Gilbert 1877) and their mode of emplacement and magma dynamics are generally understood (Corry 1988; Pollard & Johnson 1973; Johnson & Pollard 1973). Additional field based studies, however, are needed to test competing models for specific laccoliths because a single emplacement model for all laccoliths does not seem warranted given the complexities associated with their emplacement dynamics. Conventionally, structural studies of laccoliths and of plutons in general involve field and laboratory measurement of their mineral phase arrangement. The shape, orientation and spatial distribution of the mineral phases define a fabric, and the kinematic condition of its formation can be inferred from their arrangement. During magmatic flow, the early-formed crystals rotate and undergo relative translations, and they may provide indirect evidence of magmatic flow path. In this study, we use the orientation of the magnetic fabric to infer magma flow patterns during emplacement of a shallow level laccolith. Foliation and lineation orientations determined in this manner can then be used to understand better the emplacement path of intrusions with different geometries (Hutton 1988; Pitcher 1993; Bouchez 1997). Magmatic fabrics are thought to form after ascent and before final crystallization of the magma (Cruden 1990; John & Blundy 1993; Clemens *et al.* 1997). As early ascent structures are likely obliterated during magmatic flow, only the final stages of magma flow can be inferred from petrofabric studies (Hutton 1992; Bouchez 1997; Vigneresse & Clemens 2000). Although important information can be obtained from structural studies, they have typically been limited in their effectiveness in that magmatic fabrics tend to be only weakly developed and difficult to identify and measure in the field. Recently, studies aimed at describing and interpreting the internal fabrics of plutons have been greatly advanced by the application of the low-field anisotropy of magnetic susceptibility (AMS) technique (Balsey & Buddington 1960; King 1966; Heller 1973; Borradaile 1980; Hrouda 1982; Guillet *et al.* 1983; Bouchez *et al.* 1990; Archanjo *et al.* 1992; Benn *et al.* 1993; Bouillin *et al.* 1993). The AMS method can detect anisotropies of only a few per cent and, therefore, the technique has proved to be a powerful method to complement field studies of plutons (Bouchez 1997).

In this contribution, we report AMS and palaeomagnetic data from the Stoddard

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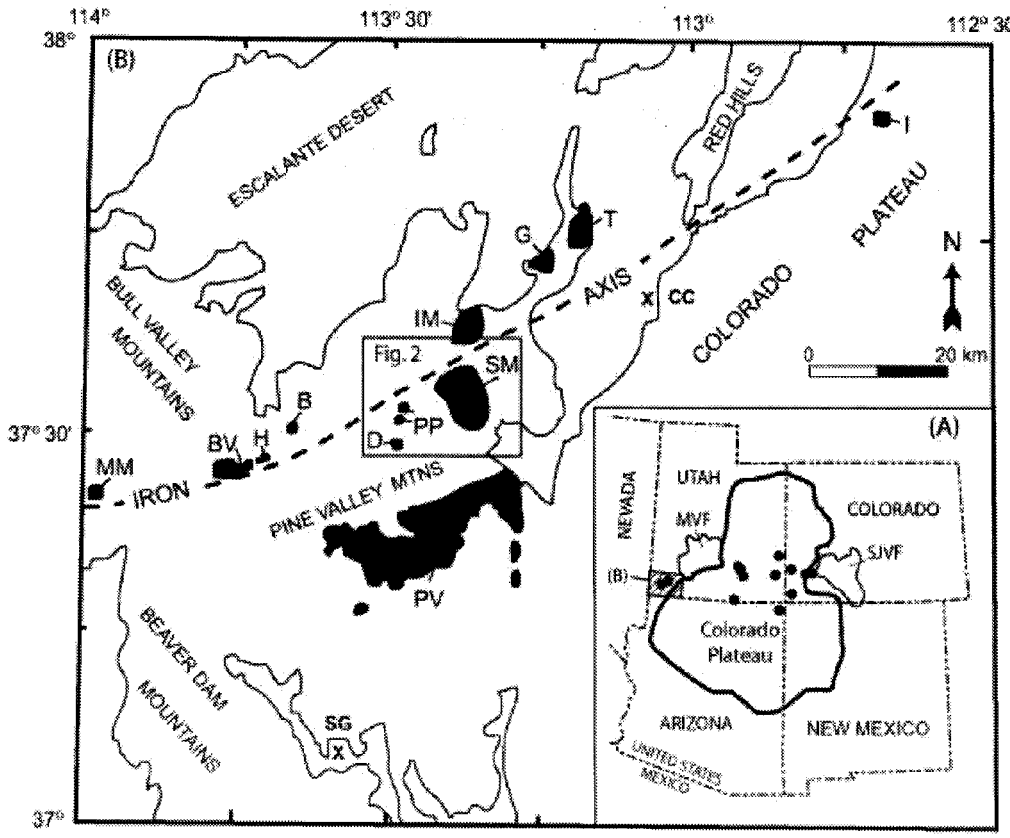


Fig. 1. (**a**) Selected igneous features of the Colorado Plateau. Black circles represent Colorado Plateau laccolithic centres; MVG – Marysvale volcanic field, and SJVF – San Juan volcanic field. Modified from Mutschler *et al.* (1998). (**b**) Index map of southwest Utah showing intrusions of the Iron Axis: B – Big Mountain; BV – Bull Valley; D – The Dairy; G – Granite Mountain; H – Hardscrabble Hollow; I – Iron Peak; IM – Iron Mountain; LP – Lookout Point; MM – Mineral Mountain; PP – Pinto Peak; PV – Pine Valley; SM – Stoddard Mountain: T – Three Peaks. Towns: CC – Cedar City; SG – St. George.

Mountain laccolith, Iron Axis magmatic province, Southwestern Utah. The Iron Axis laccoliths are part of a Late Oligocene-Early Miocene hypabyssal laccolith swarm that lies within and borders the Colorado Plateau (Fig. 1a). In the Iron Axis region, mineralization associated with the emplacement of over a dozen laccoliths has resulted in significant iron-ore production (Mackin 1968). The Stoddard Mountain intrusion, one of the largest and best exposed of these laccoliths, is an example of a granitoid in which the AMS fabrics are well defined. Its relation to the surrounding country rock and to other smaller nearby laccoliths is well understood through detailed mapping (Hacker 1998; Rowley *et al.* 2004 in press). The magnetic fabrics in the Stoddard Mountain laccolith were systematically mapped on a relatively tight sampling grid (76 sites distributed over the 54 km^2 map area) using low-field anisotropy of magnetic susceptibility (AMS). The magnetic susceptibility, degree of magnetic anisotropy, and the shapes of the AMS ellipsoids are homogeneous at the outcrop scale (10 m^2) and, on the laccolith scale, they define a pattern that is interpreted to reflect magma flow during emplacement. The remanence directions recorded in the main body of the laccolith are well grouped ($D = 351.1°$, $I = 57.8°$, $\alpha_{95} = 3.7°$, $k = 82.2$) and only slightly discordant to a Late Miocene expected field direction (358°, 58°). Sites in chill zone rocks collected throughout the perimeter of the laccolith, however, yield a magnetization of south declination and very shallow inclination.

AMS studies on other Colorado Plateau laccoliths are beginning to provide important information about magma flow patterns and magma dynamics (Horsman & Tikoff 2002). Whereas many of the Colorado Plateau laccoliths appear to have been fed from below by a central axial feeder system (Jackson & Pollard 1988; Ross 1998), our results from the Stoddard Mountain laccolith are more consistent with magma emplacement from a shallow level sill that spread laterally from the west to east. These results, used together with recent mapping, allow us to document the mode of laccolith emplacement and to assess the influence of surrounding geologic features on its development. This study may have important implications, specifically, for testing hypotheses regarding the formation of laccolith related ore-deposits in the Iron Axis region (Mackin & Ingerson 1960), as well as for better understanding the kinematics of shallow level laccolith emplacement in general.

Geological setting

The Iron Axis laccolith swarm consists of a series of Early Miocene calc-alkaline hypabyssal laccoliths and associated volcanic rocks located just west of the present Colorado Plateau in Southwestern Utah (Fig. 1b). The Iron Axis igneous rocks are part of the general Mid-Cenozoic calc-alkaline igneous sequence that spans much of the western United States. This magmatism is generally considered coincident with oblique convergence during subduction of oceanic lithosphere beneath western North America that produced large fluxes of mantle-derived mafic magma injected into the overlying continental lithosphere (Johnson 1991; Nelson & Davidson 1998; Rowley *et al.* 1998).

With the onset of calc-alkalic volcanism, prior to Iron Axis magmatism that produced laccoliths, a sequence of regional Oligocene and Miocene calc-alkaline andesite to rhyolite ash-flow tuffs of the Wah Wah Springs Formation (30 Ma), Isom Formation (27 Ma) and Quichapa Group (24 to 22.5 Ma) were spread over the area from sources to NW and west (mostly from the Indian Peak and Caliente caldera complexes; see Best *et al.* 1989 and Rowley *et al.* 1995). This sequence of pre-Iron Axis volcanic rocks overlies fluvial and lacustrine sedimentary rocks of the Upper Palaeocene-Oligocene Claron Formation, that in turn, unconformably overlie Cretaceous and Jurassic sedimentary rocks deformed during the Sevier orogeny (Late Cretaceous to Early Tertiary) that produced east and SE verging thrusts.

During Iron Axis magmatic activity, ascending magma from an inferred batholith complex intruded along one or more NE-striking Sevier-age thrust faults before being emplaced as bulbous laccoliths within Mesozoic and Tertiary sedimentary strata (Mackin 1960; Blank & Mackin 1967; Blank *et al.* 1992; Rowley *et al.* 1998; Rowley 1998). More than a dozen exposed intrusions have been mapped within the magmatic province and others are inferred from structures in the cover-rock and aeromagnetic signatures, or are known from drilling and seismic data. Thus, the intrusions are distributed along a NE-trending zone that follows the trend of the Sevier orogenic front (Fig. 1b). These intrusions were forcibly emplaced into sedimentary rocks at depths ranging mostly between 2.5 and 0.25 km from the surface and deformed their roofs by upward folding and faulting (Corry 1988; Johnson & Pollard 1973). The intrusions in the eastern Bull Valley Mountains (Bull Valley, Hardscrabble Hollow and Big Mountain) and the Iron Springs mining district (Iron Mountain, Granite Mountain and Three Peaks) intruded into limestone strata of the Jurassic Carmel Formation and produced major replacement magnetite and hematite ore bodies. Only three plutons in the central part of the Iron Axis Iron Springs mining district were extensively mined and they constitute the largest iron ore production in the western United States (Mackin 1960; Blank & Mackin 1967; Barker 1995). Iron-rich solutions were derived from deuteric breakdown of ferromagnesian minerals in the outer selvage-joint phase of the intrusions (Mackin & Ingerson 1960; Rowley & Barker 1978; Barker 1995).

Plutons of the central part of the Iron Axis are quartz monzonite to granodiorite porphyries characterized by phenocrysts of plagioclase (andesine-labradorite), biotite, hornblende and/or pyroxene (diopsidic augite), and magnetite in a groundmass (1/3 to 1/2 total volume) of very fine-grained quartz and potassium feldspar. The only exceptions to this composition range are the Mineral Mountain intrusion at the SW end of the trend, which is a granite porphyry (Morris 1980; Adair 1986), and the Iron Peak intrusion at the NE end, which is a gabbro-diorite porphyry (Spurney 1984). The exposed plutons yield K–Ar dates and $^{40}Ar/^{39}Ar$ age spectra values of 20–22 Ma (Armstrong 1970; Hacker *et al.* 1996; McKee *et al.* 1997; Rowley *et al.*, in press) indicating a two million year period of Iron Axis magmatic activity.

Both structural relief and current topographic relief have been produced by the emplacement of the Iron Axis laccoliths. The nature of the

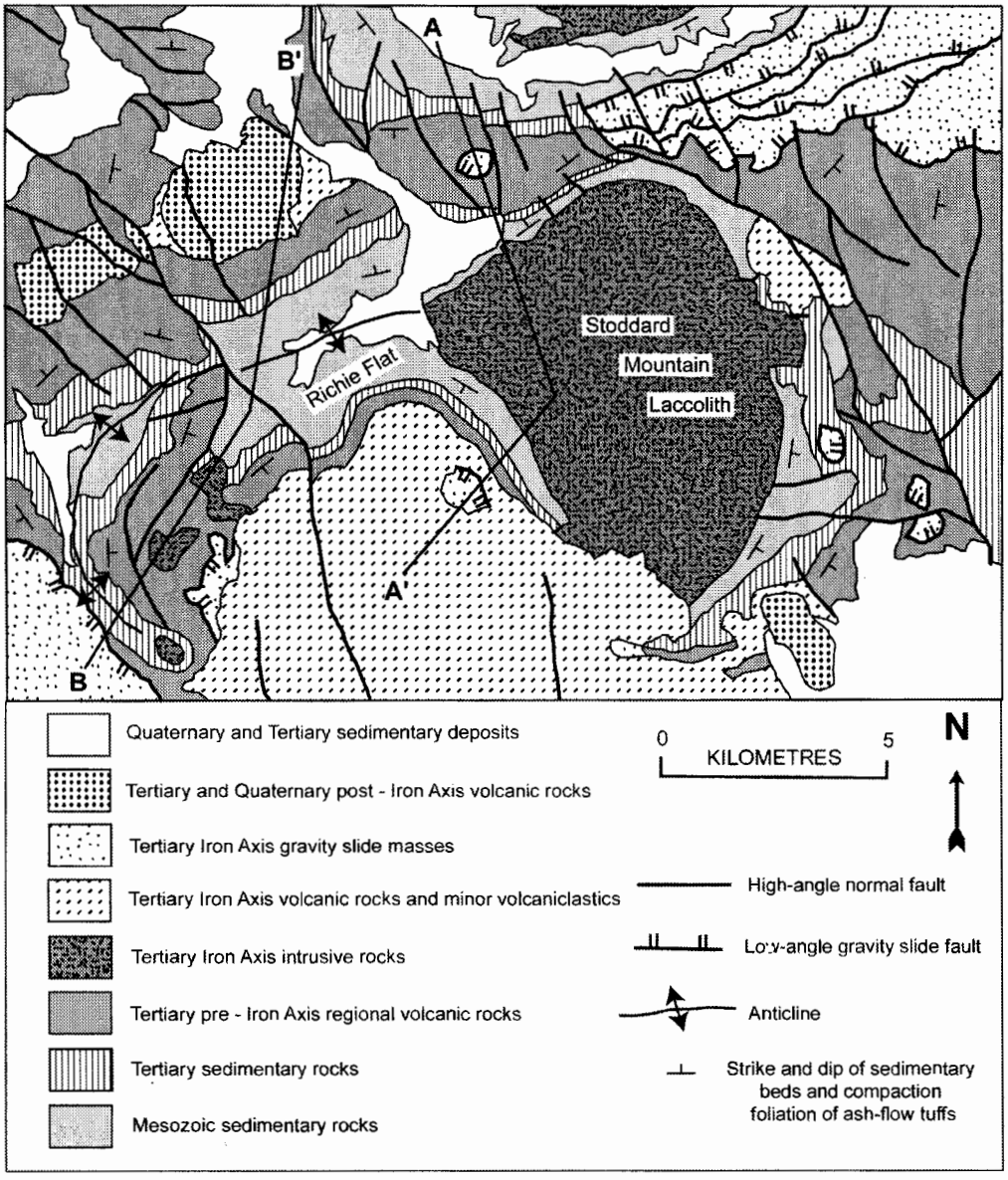


Fig. 2. (a) Simplified geological map.

igneous intrusions in the Iron Axis province has been historically controversial, in part, because the current level of erosion makes it difficult to distinguish the intrusions as stocks or laccoliths. Only the Pine Valley and Iron Peak intrusions have been eroded to expose their planar, sub-horizontal floors. However, geophysical and drill hole data, together with detailed field mapping, confirmed that the intrusions have nearly planar floors and irregular but concordant roofs (Van Kooten 1988; Hacker 1998; Hacker *et al.* 2002). Cross-sections show that most of the laccoliths have a typical plano-convex shape (Corry 1988), but are not circular in map view (Hacker 1998). During the emplacement of the laccoliths, several (e.g. Pine Valley, Bull Valley-Big Mountain, Pinto Peak, Stoddard Mountain and Iron Mountain) were partially unroofed by catastrophic gravity sliding and vented to the surface (except Iron Mountain) forming related

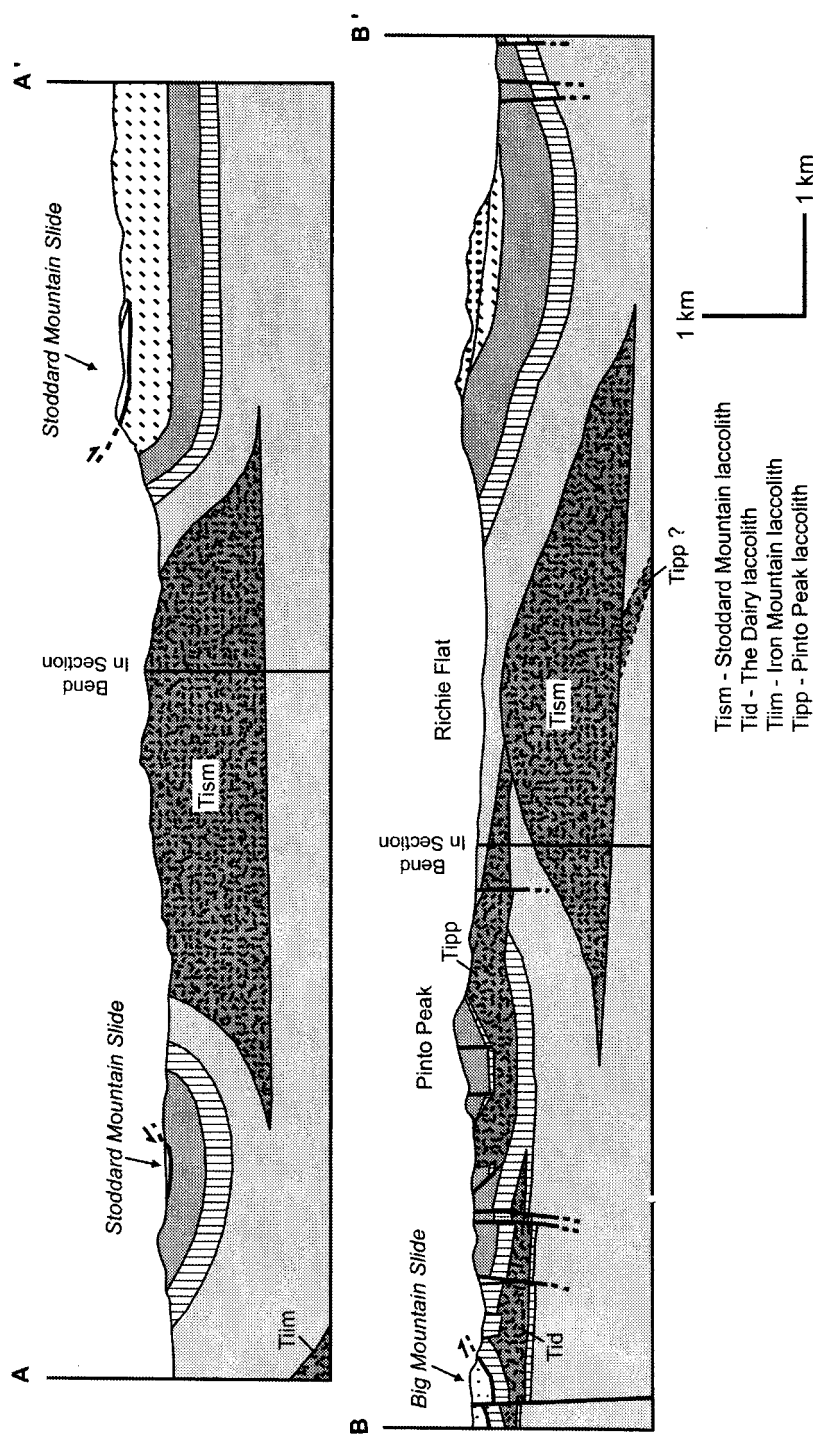


Fig. 2. (b) cross-section (A–A′ and B–B′) of the Stoddard Mountain laccolith showing its estimated extent, vertical thickness, and intrusive and extrusive features in cross-section delineates the sampling region within the laccolith.

ash-flow tuffs and minor lava flows that buried the gravity slide masses (Mackin 1960; Blank *et al.* 1992; Hacker 1995, 1998; Hacker *et al.* 2002) (Fig. 2).

Field relations

The Stoddard Mountain Laccolith, which underlies the Stoddard Mountain area in the northern Pine Valley Mountains, is one of the quartz monzonite porphyry laccoliths of the central part of the Iron Axis (Fig. 1b). The exposed part of the laccolith is roughly elliptical in shape, about 9 km long N–S and 6 km wide E–W, and about 54 km^2 in area. The unexposed part of the laccolith extends westward at least another 5 km beneath the Richie Flat anticline (Fig. 2a) as delineated by drill cores (Cook 1957). It intruded clastic sedimentary rocks of the Cretaceous Iron Springs Formation, which, along with Tertiary volcanic rocks, dip steeply away from the exposed part of the intrusion and more gently over the Richie Flat anticline. At the NE margin of the laccolith the country rock is steeply overturned. Although erosion has partially removed some of the country rocks to expose the upper crest and flank of the intrusion (the laccolith is well exposed over a topographic relief of about 700 m), the floor is not exposed. Based on geological mapping and cross-section construction, the intrusion has a remaining thickness of about 1.5 km where exposed and 1.0 km beneath Richie Flat (Fig. 2b).

The laccolith consists of homogeneous medium-grained quartz monzonite porphyry forming two gradational phases. Marking the outermost 20 to 60 m of the intrusion is a chilled margin zone consisting of fresh, resistant, light-grey, pink and light yellow porphyry. This zone consists of about 45% phenocrysts (approximately 29% plagioclase, 2% clinopyroxene, 1% biotite, trace amphibole, 1% Fe–Ti oxides, and 14% aggregates of plagioclase, sanidine, clinopyroxene, chlorite, and calcite as pseudomorphous after amphibole and biotite phenocrysts) in a fine granular groundmass of quartz and alkali feldspar (Barker 1995) (Fig. 3). The phenocrysts reach lengths of 8 mm and show no visible preferred orientation. The interior zone, which makes up most of the intrusion, consists of the same minerals as the peripheral chill zone except that the groundmass is slightly coarser (medium grained) and the minerals are more altered (Fig. 3). This zone contains abundant irregular miarolitic cavities, consistent with its shallow emplacement depth.

The intrusion post-dates deposition of the 22 Ma Harmony Hill tuff of the Quichipa Group and its age is approximated by a biotite K–Ar age of 21.5 ± 0.9 Ma (Mckee *et al.* 1997) and a $^{40}Ar/^{39}Ar$ age determination of 21.86 ± 0.09 Ma (Rowley *et al.* 2003, in press). Field evidence from the surrounding laccoliths show that the Pinto Peak and Iron Mountain laccoliths were emplaced concurrently; prior to Stoddard Mountain emplacement (Hacker 1998). Both of these intrusions produced gravity slides, and the Pinto Peak intrusion vented a thick sequence of ash-flow tuff and lava flows (Fig. 2). Stoddard Mountain and The Dairy laccoliths are interpreted as coeval intrusions, because they both deformed volcanic units that vented from the Pinto Peak laccolith. The Stoddard Mountain laccolith also vented

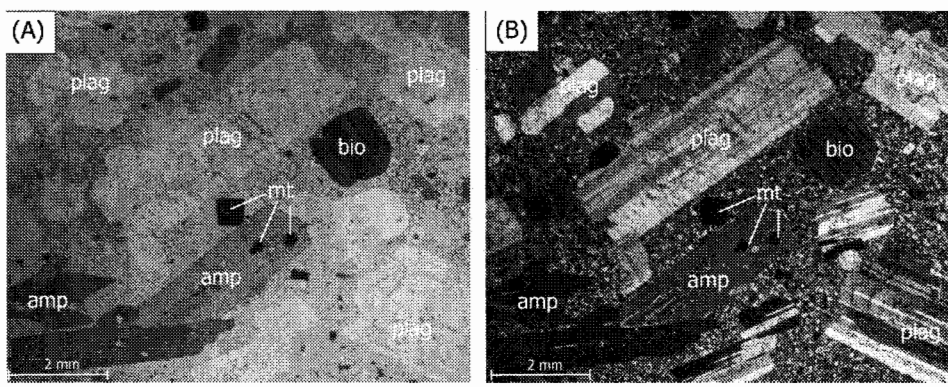


Fig. 3. Thin section of representative texture from chilled zone rocks (**a**) plane light (**b**) polarized light. Inner zone rocks (not shown) have a similar texture but the groundmass and Fe-Mg silicates are more altered. mt, magnetite; plag, plagioclase feldspar; bio, biotite; hbd, hornblende.

ash-flow tuffs and lava flows following several gravity sliding events from its flank. At least two vent areas are interpreted along its flank, one on the east side where the intrusion bulges out from the main pluton (Fig. 2), and on the SW side where lava flows grade downward into the intrusion (Hacker 1998; Rowley *et al.*, in review).

Recent mapping of the Stoddard Mountain and The Dairy laccoliths (Hacker 1995, 1998; Hacker *et al.* 2002) suggests that the intrusions were emplaced as sills, which spread laterally for several kilometres before inflating to laccolithic shapes (Johnson & Pollard 1973). This is based on field relations of intrusive anticline fold axes NE (Richie Flat) and west of the Pinto Peak laccolith that have a characteristic arcuate trend wrapping around the Pinto Peak laccolith and its volcanic pile of over 500 m of ash-flow tuffs and lavas lying E–SE of the Pinto Peak laccolith (see Fig. 2a). The intrusive cored anticlines clearly deformed part of the Pinto Peak intrusive-volcanic complex (Fig. 2a), but did so only around the thinner edges of the pile. Hacker (1998) proposed that magma of both Stoddard Mountain and The Dairy laccoliths likely emanated from a common buried conduit north of Pinto Peak. Finding the area to the south blocked by the older Pinto Peak laccolith, the magma migrated to the east (for Stoddard Mountain) and west (for The Dairy). The sills, emplaced at about 1.2 km depth within the Cretaceous Iron Springs Formation, are inferred to have spread laterally, east and west, for several kilometres around the northern edge of the thicker Pinto Peak intrusive-volcanic complex. As The Dairy sill migrated westward around the intrusive-volcanic complex, it turned southward and stepped up to a higher structural level within the Tertiary Claron Formation (Fig. 2a). As magma flowed east, beneath Richie Flat, it gently deformed the overlying country rock to produce the Richie Flat anticline. The Stoddard Mountain sill continued to migrate to the east past the intrusive-volcanic barrier, then turned southward before inflating fully. Thus, it appears that differences in lithostatic pressure caused by thickness variations of the Pinto Peak intrusive-volcanic overburden were influential in the lateral magma emplacement of both the Dairy and Stoddard Mountain laccoliths.

The major focus of this research is to test the proposed lateral flow pattern hypothesis for development of the Stoddard Mountain laccolith. Because the quartz monzonite does not display any discernable mineral preferred orientations on the outcrops or in thin sections, we undertook a detailed AMS fabric study to provide flow-direction data for the laccolith.

Sampling and Analytical Methods

Sampling methods

We confined our sampling to the lower 250 m of the exposed laccolith, in part because of accessibility limitations, but also in an effort to sample a relatively narrow magma flow level or zone. Thirty-two palaeomagnetic sites were sampled with 12 sites in the chilled zone and 20 sites in the inner zone. Seventy-six AMS stations were sampled in a relatively tight grid network over the 54 km^2 exposure with sampling stations spaced at about 1 km to 0.5 km (Fig. 4). At either a palaeomagnetic site or an AMS station, from 4 to 10 independent samples (typically six) were drilled over an area of about 25 m^2. We observed a relatively high dispersion in AMS data, at the station level, between samples in the Stoddard Mountain intrusion, which is common in granites (Rochette *et al.* 1992; Tarling & Hrouda 1993). Analysis of six or more samples per station (with at least 2 specimens per sample) yielded a statistically robust estimate of the orientations of each of the principal susceptibility axes and/or a planar fabric. In most cases, the dispersion between specimens from the same sample was low (k greater than 50°), however, at several stations, station-level dispersion was high (k less than 15°).

Analytical methods

Palaeomagnetism

Oriented samples from all sites were collected using a portable, gasoline-powered drill with a non-magnetic, diamond-tipped drill bit. Each sample was oriented always using both solar and magnetic compasses. Each core sample was cut into 2.2 × 2.5 cm right cylinder specimens, using a diamond tipped, non-magnetic saw blade. Up to four specimens per sample were obtained. Remanent magnetizations of all samples were measured using a static, three-axis 2G Enterprises magnetometer with an integrated alternating field (AF) demagnetizing unit. Specimens were progressively AF demagnetized, typically in 15 to 20 steps to a maximum field of 140 mT. Samples of high coercivity were further treated with thermal demagnetization until less than 10% of NRM remained; typically, up to 615–635 °C depending on rock type and behaviour. Thermal demagnetization of replicate specimens, to compare with AF behaviour, was conducted using a Schonstedt TSD-1 thermal demagnetizer, with an ambient field of less than 20 nT in the heating chamber and 3 nT in the

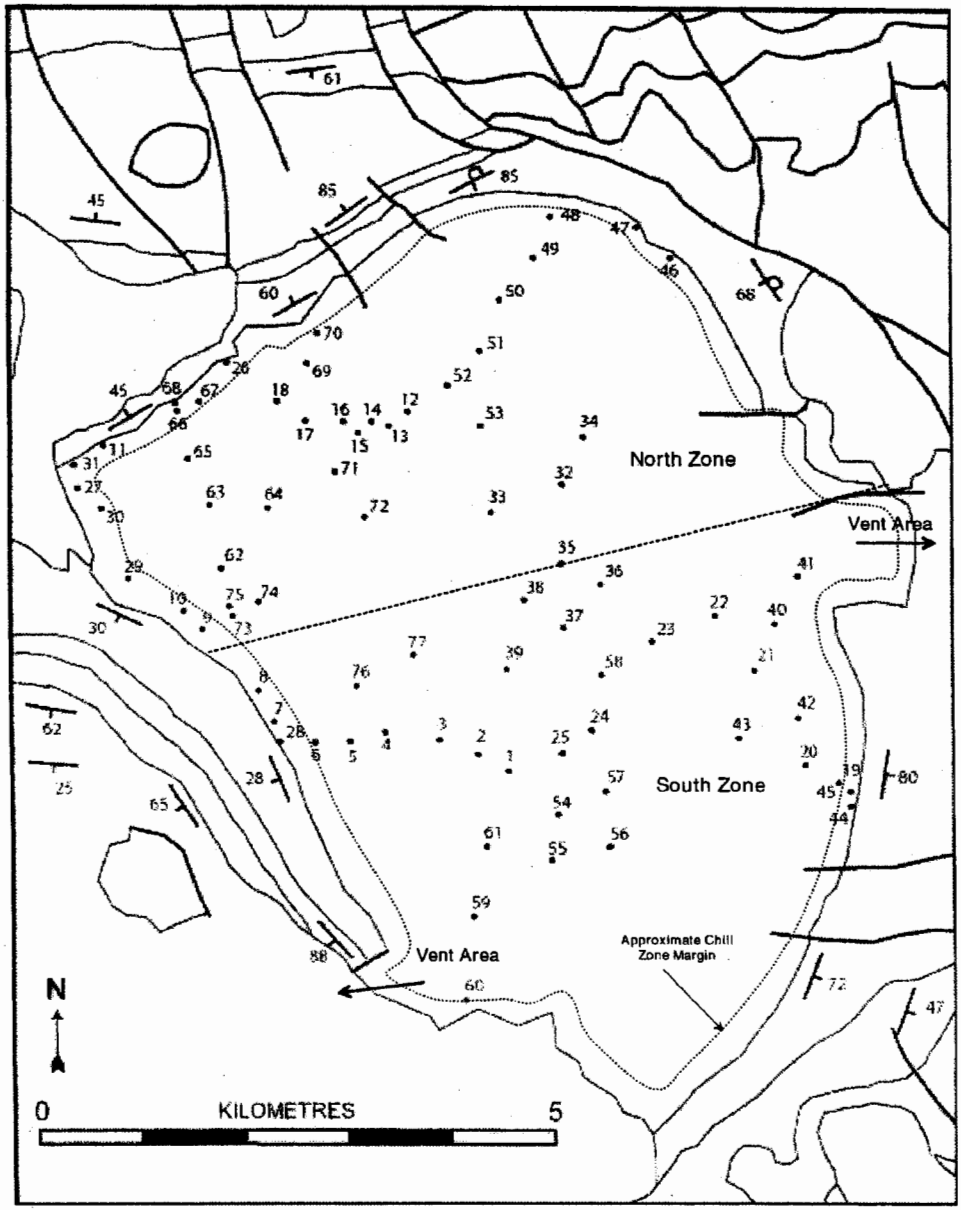


Fig. 4. Outline of the Stoddard Mountain laccolith. Numbered sample locations indicated by black dot correspond to locations numbered in Table 1 and Table 2. Dashed line: boundary between northern and southern zones.

cooling chamber. Principal component analysis (PCA; Kirshvink 1980) was used to determine the best-fit line through selected demagnetization data points for each sample. For most samples, a single best-fit line could be fit to the demagnetization data points. Best-fit magnetization vectors, typically, involved 10 to 15 data points. Sample maximum angular deviation (MAD) values for linear data, typically, were less than 4°, but ranged from less than 1° to 5°. For less than 10% of the demagnetization results, it was necessary to anchor the magnetization vector to the origin. Data from individual specimens were considered outliers and rejected from the site

mean calculation if the angular distance between the specimen direction and the estimated site mean direction was greater than 15°.

Rock magnetism

To characterize the magnetic mineralogy of the Stoddard Mountain intrusion, standard rock magnetic experiments were conducted with the goal of identifying the magnetic phases carrying the remanence and the overall ability of these rocks to faithfully record an ambient field. The tests include AF demagnetization of anhysteretic remanent magnetization (ARM), d.c. acquisition of a saturation isothermal remanent magnetization (IRM), d.c. demagnetization of the saturation IRM to yield a backfield IRM (coercivity of remanence), AF demagnetization of saturation isothermal remanent magnetization (SIRM), bulk susceptibility in varying fields between 3 A/m and 450 A/m in 21steps, monitoring bulk susceptibility changes during thermal demagnetization, continuous susceptibility vs. temperature experiments, thermal demagnetization of three-component orthogonal IRM (Lowrie 1990), and analysis of low temperature magnetic susceptibility.

Anhysteretic remanent magnetization (ARM) was imparted in a d.c. field of 1 mT and a peak AF field of 95 mT, and then AF demagnetized in about 13 steps to 95 mT. An IRM acquisition experiment was then conducted where the samples are treated in a stepwise fashion to higher induction fields until they reach a saturation magnetization. A backfield-IRM experiment was then conducted by applying specimens to a 1.33 T isothermal induction field along the $+z$ axis and treating the specimens to increasing induction fields along the $-z$ axis until the $+Z$-remanence was reduced to zero (i.e. the sign flipped). Finally, the samples were again saturated in a 1.33 T induction field and then AF demagnetized in about 13 steps to 95 mT. For the thermal demagnetization of three-component IRM, we used fields of 3.0, 0.3 and 0.03 T, imparted using an impulse magnet. Analysis of the low-temperature magnetic susceptibility involved cooling samples to 77 K and continuously measuring susceptibility during heating to 298 K to obtain a temperature vs. susceptibility curve.

Anisotropy of magnetic susceptibility

The magnetic susceptibility (K) is defined by $M = KH$, where M is the induced magnetization of the material and H is the inducing magnetic field. The volumetric susceptibility (K) is dimensionless, because both M and H are expressed in A/m. The output of an AMS measurement of one rock specimen is an ellipsoid of magnetic susceptibility defined by the length and orientation of its three principal axes, $K_1 \geq K_2 \geq K_3$, which are the three eigenvectors of the susceptibility tensor. Depending on magnetic mineralogy, granite rocks may be generally characterized as Mt (magnetite)-bearing or Fe–Mg silicate dominated. In Fe–Mg silicate dominated granites, the silicates such as biotite or amphibole largely control AMS. In Mt-bearing granites, the contribution of Fe–Mg silicates is negligible with respect to the magnetite contribution because of the high intrinsic magnetic susceptibility of magnetite. The magnetic fabric, in that case, results primarily from the shape anisotropy of the grains of magnetite. This effect was postulated by Stacey (1960), and examined by microscopic observations on various rock types (Khan 1962; Uyeda *et al.* 1963) including granites (Ellwood and Whitney 1980; Gregoire *et al.* 1995). The AMS parameters are at times difficult to interpret in Mt-bearing granites, because of the unknown relationship between the fabric of magnetite (low volume %) and the fabrics of the volumetrically dominant phases (feldspars, amphibole, biotite). The usual observation, when magnetite crystallizes in association with K-feldspar, hornblende and biotite, is that the shape of the magnetite grains is controlled by the crystalline faces of those minerals (Hrouda *et al.* 1971). The magnetite grains use the Fe–Mg silicates as a 'template' during grain growth; therefore, the magnetic fabric is a good image of the rock fabric. This 'template' behaviour has been demonstrated on ferromagnetic granites with quantitative image analysis of thin sections (Cruden & Launeau 1994; Archanjo *et al.* 1995). Reflected light petrography confirms the presence of magnetite in association with Fe-bearing silicate phases in the Stoddard Mountain laccolith (Fig. 3).

We measured the AMS of 959 specimens prepared from samples collected at 76 sites distributed across the intrusion (Fig. 4). AMS measurements were preformed on a Kappabridge KLY-4S, operating at low alternating field 3.7×10^{-5} T (300 A/m) at 875 Hz. These measurements give the orientation and magnitude of the K_1, K_2, K_3 axes of the ellipsoid; maximum, intermediate, minimum, respectively. The magnetic lineation, K_1, and the magnetic foliation, perpendicular to K_3, characterize the fabric (Borradaile 1980; Hrouda 1982; Bouchez 1997, 2000). The relationships between tensor axes, normalized by the method of Jelínek (1977), were studied using the anisotropy degree, P', and the shape parameter, T. T varies between -1 (prolate ellipsoids) and $+1$ (oblate

ellipsoids) (Jelínek 1981). The orientation data (summarized in Table 2) are the foliation, which is the plane (K_1, K_2) perpendicular to the smallest axis (K_3) of the ellipsoid, and the lineation, which is the long axis of the AMS ellipsoid (K_1). The average, or bulk, susceptibility is given by $K = (K_1 + K_2 + K_3)/3$.

Palaeomagnetism

General demagnetization behaviour

Of the 32 sample sites, 30 sites yield acceptable results. The two excluded sites (SM8 and SM20) had a high dispersion ($\alpha_{95} > 15°$, $k < 15$) and did not yield interpretable results. These sites are not discussed further. We note that at many of the sites only 6 samples were collected, because the principal reason for carrying out this study was to obtain and evaluate AMS data to infer emplacement fabrics. With a more conventional number of samples (e.g. seven to ten) at each site, the confidence and precision estimates would probably improve. A structural correction is not available for any parts of the laccolith. Therefore, all results are reported in '*in situ*' (geographic) coordinates. In general, 11 sites in chill zone rocks, distributed around the perimeter of the intrusion, yield south declination and shallow to moderate negative and positive inclination mean directions; and 19 sites in the inner zone rocks yield north to NW declination, moderate positive inclination mean directions. Two sites in the inner zone rocks yielded south declination, moderate steep inclination means; these sites are antipodal to the remaining inner zone sites and we have therefore inverted them through the origin. Overall, progressive alternating field (AF) response of these rocks is characterized by high quality results (Fig. 5). Duplicate specimens treated with thermal demagnetization yielded directional data similar to those resolved in AF demagnetization (Fig. 5). A clear difference exists in demagnetization behaviour and remanence data (see below), between chilled zone rocks and inner zone rocks. Representative demagnetization diagrams show the variation between the two rock types and the overall high quality of the demagnetization response (Fig. 5a, b).

Chill zone

Chill zone rocks show a relatively complex demagnetization behaviour characterized by superimposed magnetization components (Fig. 5b). Most specimens yielded a low coercivity N–NE declination, steep, positive inclination overprint that is readily removed by about 20 mT. The remaining component, defined by roughly 30% of the NRM intensity, decays along a near univectoral path to the origin and is well grouped at the site level with 10 to 40% of the NRM intensity remaining after AF treatment to 140 mT. Median destructive fields range from about 10 mT to 40 mT. Selected treatment of high coercivity specimens with thermal demagnetization up 635 °C did not isolate additional magnetization components (Fig. 5b).

The eleven accepted sites in the chill zone rocks yield S–SE declination, shallow negative or positive inclination magnetizations that are well grouped (eight sites with $\alpha_{95} < 15°$) at the site level. Two sites (not included with the eleven chill zone sites), collected within a gradational, transition zone between the chill and inner zones, yield south declination, moderate negative inclination mean directions (Table 1) and, because the mean directions are essentially antipodal (less than one angular standard deviation from the overall inner group mean) to the results characteristic of the inner zone, they have been inverted and included to define a grand inner zone mean. The eleven sites provide an overall, well-defined, estimated group mean ($D = 178.5°$, $I = 0.9°$, $\alpha_{95} = 10.0°$, $k = 24.1$) (Fig. 6). The result from the eleven sites lies well over 90° from a Mid-Miocene normal polarity expected field and over 45° from a reverse polarity expected field, based on the palaeomagnetic pole of Mankinen *et al.* (1987) (Fig. 6).

Inner zone

Demagnetization of inner zone rocks reveals an essentially single magnetization component that is well grouped at the site level with median destructive fields (MDF) from about 30 mT to about 80 mT (Fig. 5a). Some samples, however, also contain a superimposed random magnetization component that is readily removed by about 15 mT or by 300 °C, which we interpret to be a low-coercivity viscous overprint (VRM) (Fig. 5a). After removing the VRM, the characteristic remanent magnetization (ChRM), which we interpret as the primary thermoremanent magnetization (TRM) and of normal polarity, decays along a roughly univectoral path to the origin with less than 10% of the NRM intensity remaining after treatment in 140 mT fields or by 620 °C (Fig. 5a).

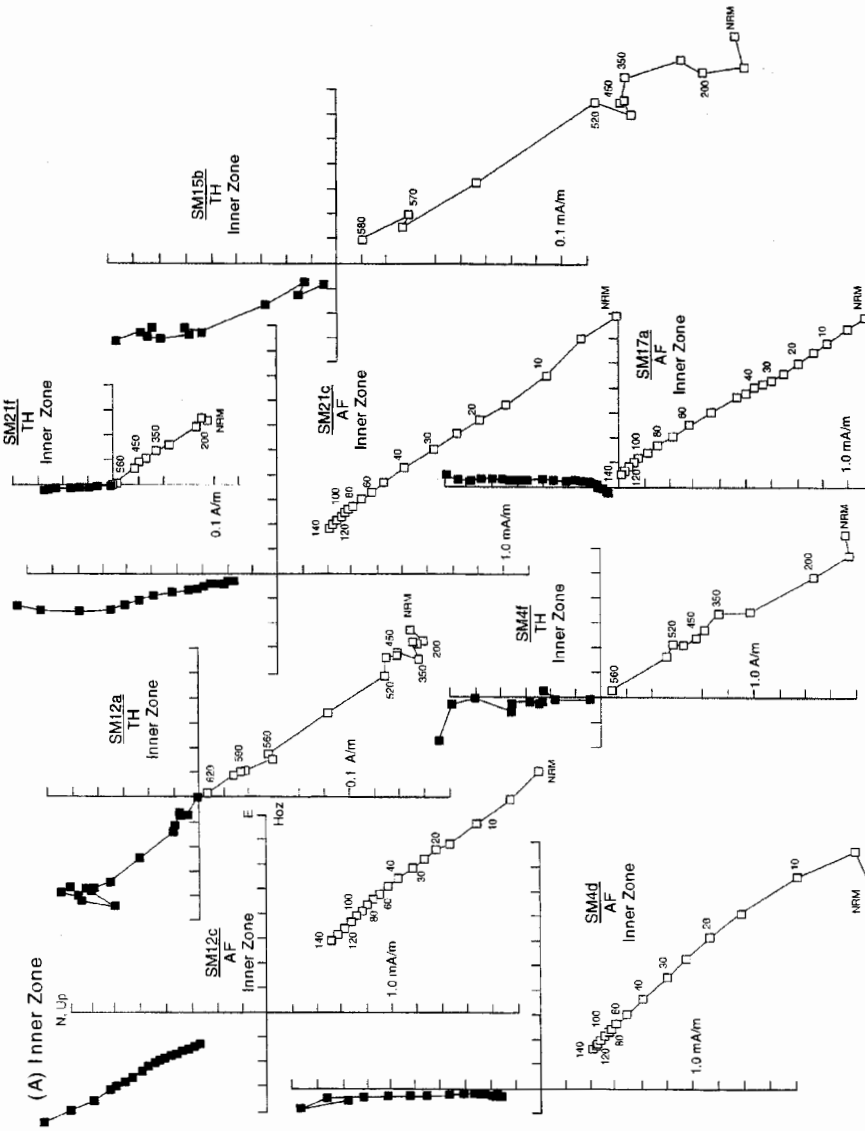


Fig. 5. Representative modified demagnetization diagrams (Zijderveld 1967; Roy & Park 1974) for rocks collected from the Stoddard Mountain laccolith (*in situ* coordinates). (**a**) Inner zone. Solid (open) symbols represent the projection onto the horizontal (true vertical) plane. AF demagnetization steps are given in milliTesla and thermal demagnetization steps in degrees Centigrade. Typically, AF and thermal demagnetization results from two specimens of the same sample are shown for comparison. Diagrams are designated by a site number (SM1), specimen letter (-b), method of treatment (AF or TH). Magnetization intensity (A/m), is shown along one axis for each sample.

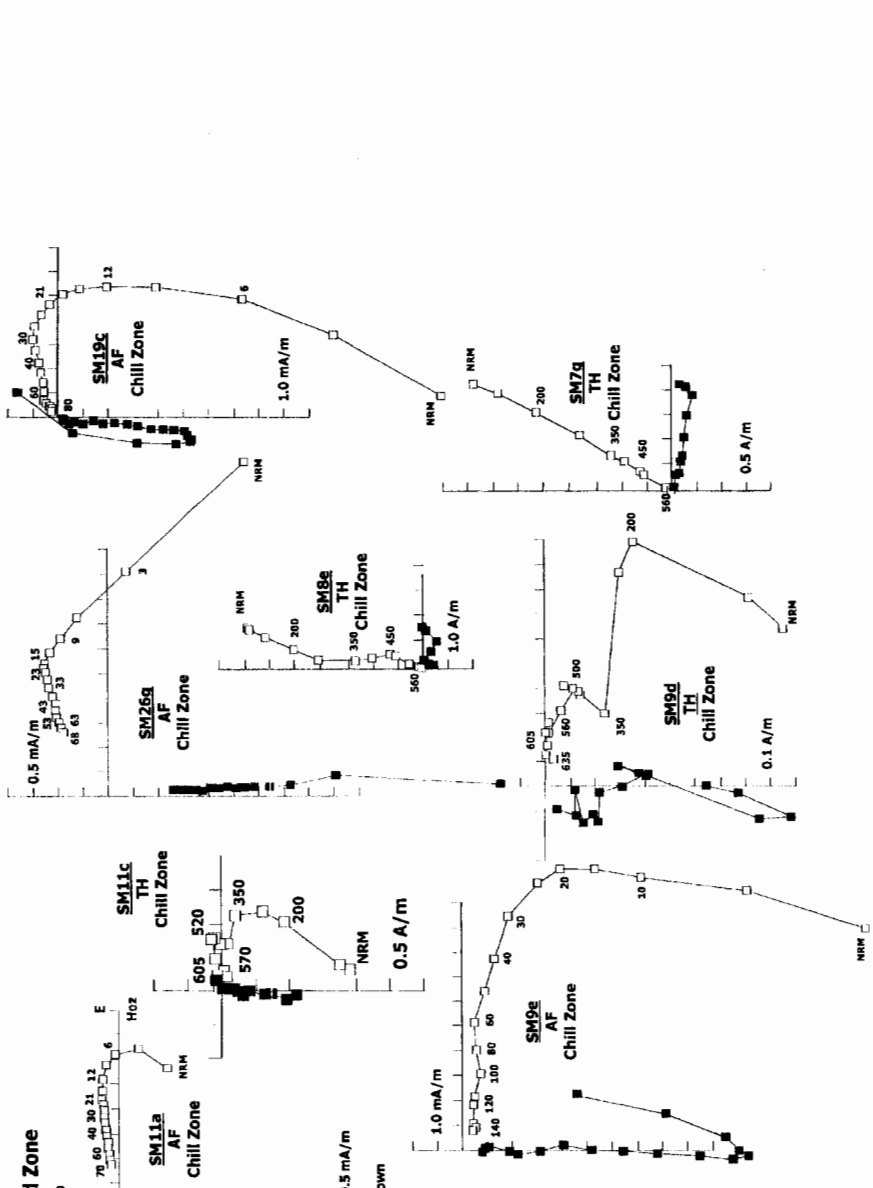


Fig. 5. Representative modified demagnetization diagrams (Zijderveld 1967; Roy & Park 1974) for rocks collected from the Stoddard Mountain laccolith (*in situ* coordinates). (**b**) Chilled zone. Solid (open) symbols represent the projection onto the horizontal (true vertical) plane. AF demagnetization steps are given in milliTesla and thermal demagnetization steps in degrees Centigrade. Typically, AF and thermal demagnetization results from two specimens of the same sample are shown for comparison. Diagrams are designated by a site number (SM1), specimen letter (-b), method of treatment (AF or TH). Magnetization intensity (A/m), is shown along one axis for each sample.

Table 1. *Anisotropy of magnetic susceptibility data from the Stoddard Mountain laccolith, in geographic (in situ) coordinates. (For symbol description, see explanation attached to data table)*

Site	No		N	Km	K1	K3	L	F	P	Pj	T	Shape	Rk Type
SM-1	6	22	18	1.73	171/16	262/3	1.009	1.001	1.009	1.010	−0.870	P	Inner
SM-2	7	23	19	2.77	181/2	287/84	1.005	1.002	1.007	1.007	−0.310	P	Inner
SM-3*	7	20	10	1.31	73/38	184/25	1.006	1.003	1.008	1.009	−0.354	P	Inner
SM-4	6	19	13	1.70	175/1	84/34	1.009	1.003	1.012	1.012	−0.469	P	Inner
SM-5	6	18	14	26.19	144/25	280/57	1.007	1.013	1.021	1.021	0.283	O	Inner
SM-6	6	18	12	14.01	123/14	353/69	1.012	1.015	1.027	1.027	0.089	O	
SM-7	6	14	14	20.76	303/8	44/52	1.003	1.012	1.015	1.015	0.611	O	Chilled
SM-8	7	14	11	21.32	148/16	38/50	1.008	1.014	1.022	1.022	0.254	O	Chilled
SM-9	6	12	8	14.81	311/2	47/73	1.005	1.012	1.017	1.018	0.435	O	Chilled
SM-10	6	13	7	15.39	160/7	292/80	1.013	1.006	1.019	1.019	−0.326	P	Chilled
SM-11	6	13	12	23.70	5/42	153/43	1.004	1.013	1.016	1.017	0.543	O	Chilled
SM-12	6	12	9	1.95	130/8	320/82	1.002	1.009	1.011	1.011	0.637	O	Inner
SM-13	6	12	8	2.04	168/1	77/52	1.004	1.006	1.009	1.009	0.205	O	Inner
SM-14	6	11	9	1.59	225/16	75/71	1.003	1.003	1.007	1.007	−0.044	P	Inner
SM-15*	6	12	10	6.96	50/26	145/10	1.005	1.007	1.012	1.012	0.131	O	Inner
SM-16	4	12	6	6.59	20/1	114/71	1.009	1.003	1.012	1.012	−0.492	P	Inner
SM-17	6	12	9	1.89	63/4	155/32	1.002	1.004	1.006	1.006	0.218	O	Inner
SM-18	7	14	11	5.31	49/38	154/19	1.008	1.002	1.010	1.011	−0.575	P	Inner
SM-19	6	12	9	20.53	195/49	91/11	1.009	1.003	1.012	1.012	−0.571	P	Chilled
SM-20	6	12	10	17.86	100/11	324/76	1.008	1.010	1.016	1.016	0.128	O	Inner
SM-21*	6	12	12	1.21	223/51	90/29	1.001	1.001	1.002	1.002	0.233	O	Inner
SM-22	5	13	8	10.13	29/3	297/35	1.012	1.005	1.017	1.018	−0.388	P	Inner
SM-23	6	12	10	8.20	43/7	296/67	1.007	1.006	1.013	1.013	−0.150	P	Inner
SM-24	6	12	8	31.41	193/27	330/56	1.017	1.004	1.021	1.022	−0.639	P	Inner
SM-25	7	12	8	23.07	214/4	124/9	1.008	1.008	1.016	1.016	−0.010	P	Inner
SM-26	7	14	10	18.81	3/34	171/56	1.009	1.007	1.016	1.016	−0.074	P	Chilled
SM-27	6	12	11	24.68	290/19	96/70	1.011	1.010	1.021	1.021	−0.021	P	Chilled
SM-28	5	19	8	18.83	181/28	65/39	1.014	1.013	1.027	1.027	−0.023	P	Chilled
SM-29*	5	11	7	27.25	259/13	358/34	1.007	1.004	1.011	1.001	−0.276	P	Chilled
SM-30	6	12	10	24.35	114/42	325/48	1.006	1.011	1.017	1.017	0.299	O	Chilled
SM-31	6	8	8	24.40	268/25	136/56	1.006	1.018	1.024	1.025	0.474	O	Chilled
SM-32	5	12	8	8.28	65/7	230/83	1.006	1.007	1.013	1.013	0.113	O	Inner
SM-33	4	12	10	6.23	51/25	302/36	1.008	1.008	1.016	1.016	0.003	O	Inner
SM-34*	6	12	12	30.27	11/6	202/84	1.003	1.016	1.019	1.020	0.664	O	Inner
SM-35*	5	12	10	19.44	336/4	76/71	1.008	1.004	1.013	1.013	−0.309	P	Inner
SM-36	5	12	8	17.65	329/15	166/74	1.004	1.006	1.010	1.011	0.152	O	Inner
SM-37	5	12	10	23.84	119/19	13/39	1.007	1.004	1.011	1.011	−0.310	P	Inner
SM-38	6	12	11	8.18	100/8	268/82	1.007	1.004	1.011	1.011	−0.200	P	Inner
SM-39	7	12	10	10.89	224/21	327/30	1.007	1.008	1.015	1.015	0.130	O	Inner
SM-40*													Inner
SM-41	8	13	10	2.71	70/9	256/81	1.003	1.003	1.007	1.007	−0.033	P	Inner
SM-42	6	12	12	3.57	99/20	227/59	1.001	1.007	1.008	1.008	0.736	O	Inner
SM-43	6	13	7	12.83	29/7	122/26	1.009	1.010	1.019	1.019	0.038	O	Inner
SM-44	7	11	8	23.13	353/10	89/31	1.009	1.011	1.020	1.020	0.119	O	Chilled
SM-45	8	12	12	15.06	274/64	11/4	1.008	1.005	1.014	1.014	−0.224	P	Inner
SM-46	5	12	10	21.35	131/21	253/54	1.020	1.013	1.034	1.034	−0.223	P	Chilled
SM-47	5	12	9	23.09	88/62	205/14	1.010	1.002	1.012	1.013	−0.619	P	Chilled
SM-48	8	12	12	2.37	2/5	266/51	1.003	1.015	1.017	1.019	0.692	O	Inner
SM-49	7	12	10	4.89	40/32	247/55	1.006	1.006	1.013	1.013	0.002	O	Inner
SM-50	6	12	12	18.24	70/13	286/75	1.012	1.005	1.016	1.017	−0.433	P	Inner
SM-51	7	8	7	2.04	73/59	235/30	1.008	1.008	1.016	1.016	−0.044	P	Inner
SM-52	8	12	11	13.52	52/22	211/67	1.006	1.014	1.020	1.021	0.371	O	Inner
SM-53*	6	12	9	12.58	153/8	6/81	1.003	1.015	1.018	1.019	0.667	O	Inner
SM-54	6	12	12	4.27	128/48	302/42	1.008	1.013	1.022	1.022	0.224	O	Inner
SM-55	5	12	9	7.91	191/60	313/17	1.010	1.010	1.020	1.020	−0.025	P	Inner
SM-56	6	12	11	9.09	156/59	289/22	1.006	1.006	1.012	1.012	−0.005	P	Inner
SM-57	8	12	12	6.67	128/45	356/34	1.007	1.015	1.022	1.023	0.353	O	Inner
SM-58	5	12	9	23.84	211/13	317/52	1.009	1.008	1.017	1.017	−0.047	P	Inner
SM-59	6	12	12	4.12	115/31	303/58	1.006	1.007	1.013	1.013	0.103	O	Inner

Table 1. (cont.) *Anisotropy of magnetic susceptibility data from the Stoddard Mountain laccolith, in geographic (in situ) coordinates. (For symbol description, see explanation attached to data table)*

Site	No		N	Km	K1	K3	L	F	P	Pj	T	Shape	Rk Type
SM-60	6	12	9	8.99	172/23	263/1	1.012	1.004	1.016	1.016	−0.537	P	Inner
SM-61	5	12	9	3.40	186/25	285/20	1.009	1.009	1.018	1.018	−0.050	P	Inner
SM-62	10	12	12	28.52	82/2	205/87	1.004	1.016	1.020	1.021	0.620	O	Inner
SM-63*	6	13	10	8.82	347/17	102/55	1.006	1.002	1.008	1.008	−0.515	P	Inner
SM-64*	5	12	8	11.63	185/4	67/83	1.009	1.001	1.010	1.011	−0.826	P	Inner
SM-65	5	12	8	24.00	61/36	253/54	1.006	1.009	1.015	1.015	0.248	O	Inner
SM-66	6	12	8	16.31	34/32	168/49	1.009	1.009	1.019	1.019	−0.026	P	Chilled
SM-67	8	14	13	27.53	353/9	105/67	1.006	1.014	1.020	1.020	0.355	O	Chilled
SM-69	7	14	12	2.16	295/23	152/62	1.004	1.008	1.012	1.012	0.304	O	Inner
SM-70*	6	12	8	3.19	264/35	154/26	1.012	1.002	1.014	1.015	−0.735	P	Inner
SM-71	7	13	13	4.33	38/46	258/37	1.000	1.002	1.003	1.003	0.624	O	Inner
SM-72*	8	14	11	2.19	340/16	233/47	1.004	1.001	1.005	1.005	−0.500	P	Inner
SM-73	8	9	9	23.33	257/9	124/77	1.011	1.003	1.013	1.014	−0.606	P	Inner
SM-74	11	14	14	31.03	140/20	32/41	1.013	1.007	1.020	1.020	−0.313	P	Inner
SM-75	5	7	6	25.31	3/3	94/28	1.009	1.002	1.011	1.012	−0.664	P	Inner
SM-76	6	12	10	28.20	38/14	190/74	1.014	1.012	1.026	1.026	−0.103	P	Inner
SM-77	7	12	11	3.43	163/7	67/40	1.009	1.005	1.014	1.014	−0.332	P	Inner

No, number of samples collected at each site; N, number of accepted specimens at each site, typically, one to three specimens per sample; Km, magnitude of susceptibility (in 10^{-3} SI); K1, azimuth and plunge (in degrees) of magnetic lineation; K3, azimuth and plunge (in degrees) of normal to magnetic foliation plane; L, magnetic lineation ((K1 - K2)/Km); F, magnetic foliation ((K2 − K3)/Km), P, anisotropy degree (K1/K3); Pj, magnitude of anisotropy, corrected anisotropy (Jelínek 1981); T, shape

The 19 acceptable sampling sites established yield interpretable demagnetization data of N–NW declination and moderate, positive inclination values that are well grouped (18 sites with $\alpha_{95} < 10°$) at the site level (Table 1). The 19 results are used to calculate an overall group mean ($D = 351.1°$, $I = 57.8°$, $\alpha_{95} = 3.7°$, $k = 82.2$) that is well defined and slightly discordant, in an anticlockwise sense, to the Mid-Miocene expected direction (358°, 58°), based on the palaeomagnetic pole of Mankinen *et al.* (1987) (Fig. 6).

Palaeomagnetic field tests

Contact test

Comprehensive contact tests were attempted at three locations on the margin of the intrusion in order to assess the antiquity of the magnetization. Unfortunately, the laccolith was emplaced into the Cretaceous Iron Springs Formation, a very fine- to medium-grained sandstone interbedded with black shale, and, where exposed, it intruded into the black shale layer, which prevented us from collecting samples directly adjacent to the intrusion. Therefore, an ideal contact test could not be conducted and a modified contact test was attempted. This test entailed collecting oriented samples from the sandstone as close as possible to the intrusive contact (<1 m) and then sampling away from the intrusive contact in exposed sandstone layers. All modified tests yield positive results. Sites closest to the intrusive contact (< ~1 m) appear to be thermally annealed and yield remanence directions similar to the intrusive rocks, regardless of the orientation of the Cretaceous strata. Sites located greater than 1 m from the contact yield poorly defined remanence directions and exhibit demagnetization behaviour characteristic of low-coercivity, multidomain magnetite, which is typical of the Iron Springs Formation in areas well away from the intrusion.

Chilled margin test

At the eastern margin of the intrusion, a road cut allowed for a non-standard field test to be conducted. We informally refer to this test as a chilled margin test (Fig. 7). The field test is based on the observation that the inner zone and chill zone rocks yield significantly different remanence directions. As noted above, chill zone rocks are considered to form a thin carapace to the intrusion, between 20 and 60 m thick, which is easily recognized in the field by a low degree of oxidation of Fe-bearing silicate minerals. The inner zone rocks, however, are only moderately oxidized. The nature of the test involved a traverse across the inner zone–chill zone boundary that started at the intrusive-country rock contact and ended about

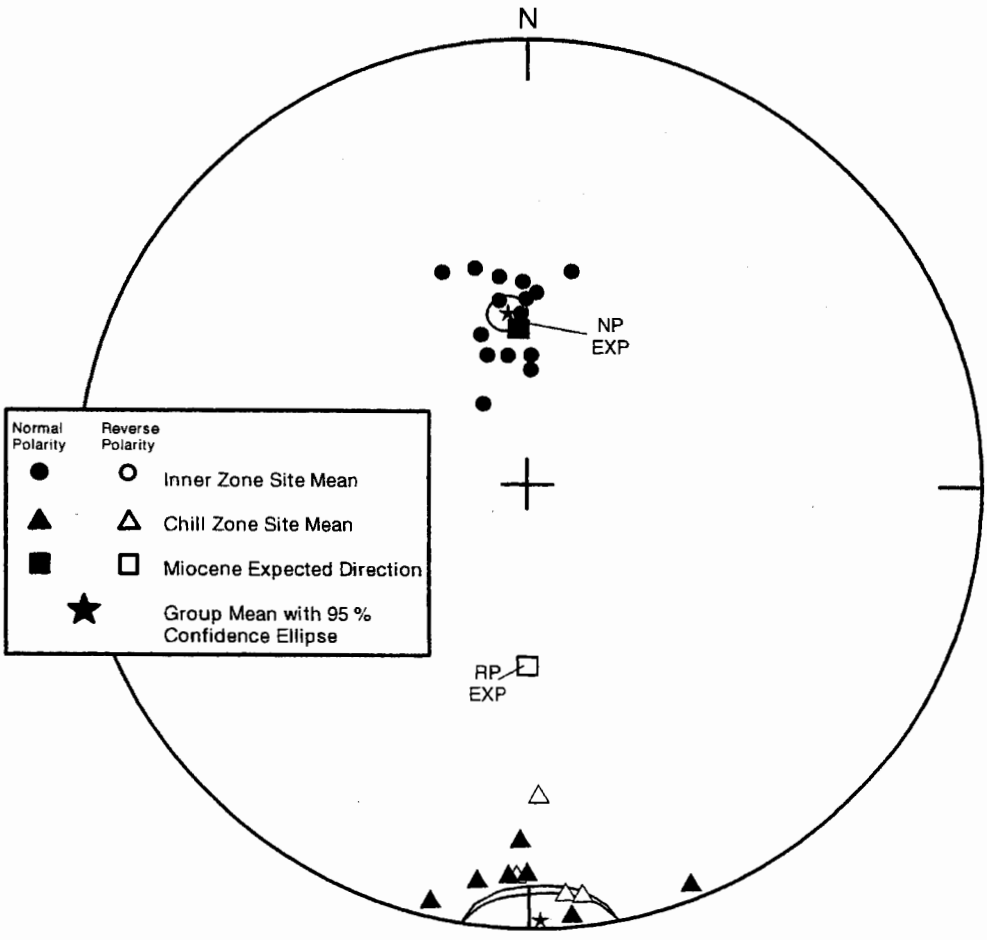


Fig. 6. Equal area projections of in situ site mean directions from the Stoddard Mountain laccolith. (lower hemisphere projection = solid circle; upper hemisphere projection = open circle; ellipse, 95% confidence ellipse; normal (reversed) polarity Late Oligocene–Early Miocene expected direction = solid (open) square)

75 m into the intrusion; a distance that should have included the chill zone–inner zone boundary (Fig. 7). A complete palaeomagnetic site was established at either end of the traverse (SM45 west side; SM44 east side) and a total of 28 samples were collected along the traverse at a spacing of about two to ten metres and, in the region where the degree of oxidation of Fe-bearing silicates increased, samples were collected every half metre. The test did not yield a 'positive' result. Based on field observation, it was anticipated that the chill zone–inner zone transition should be relatively abrupt and the south declination/very shallow inclination characteristic remanence direction of the chill zone rocks should progressively, over a short distance, change to that characteristic (north declination/moderate positive inclination) of the inner zone rocks. The directional data from the 28 specimens collected along the traverse yield consistent south-directed declinations, but a general pattern of inclination steepening from east to west into the 'main' body of the laccolith (Fig. 7). Data from the two palaeomagnetic sites also reflect this general trend (Fig. 7). Site SM44, located at the extreme eastern boundary of the laccolith, yields a south declination, shallow inclination mean ($D = 174.7°$, $I = -9.6°$) and SM45, located about 75 m west of SM44, yields a south declination, more moderate inclination mean ($D = 177.8°$, $I = -30.9°$) (Fig. 7). Based on these data, the chill zone

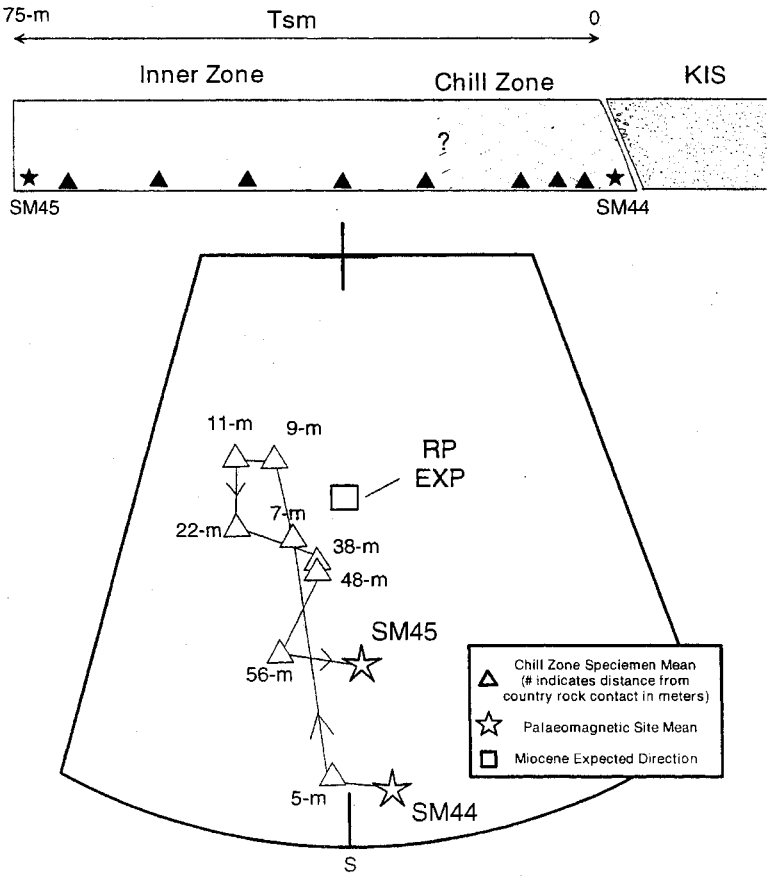


Fig. 7. Palaeomagnetic chilled margin test. Samples (indicated by black triangles in upper diagram) collected at increasing distances from the intrusive–country rock contact yield progressively steeper inclination values and basically form a loop that trends towards the expected reversed polarity field direction (lower diagram). Lower diagram partial lower hemisphere equal area projection.

boundary, defined by the alteration degree of the oxide mineral phases, does not correspond with the remanence-defined boundary. It appears that the remanence-defined inner zone–chill zone boundary is characterized by intermediate remanence directions.

Rock magnetism and AMS results

Rock magnetic properties

Standard rock magnetic tests indicate that the remanence and the AMS fabrics in the Stoddard Mountain laccolith are likely defined by a ferromagnetic mineral phase, probably multi-domain (MD) to pseudo-single domain (PSD) magnetite, although some variations were observed. Analysis of the low temperature magnetic susceptibility reveals that paramagnetic and ferromagnetic minerals both contribute to the susceptibility, and heating curves of most samples are the result of the superposition of the paramagnetic and ferromagnetic effect (Richter & van der Pluijm 1994), although some samples exhibit temperature independent behaviour, indicating the presence of single domain (SD) magnetite (Fig. 8). Susceptibility magnitudes for paramagnetic mineral phases never exceed 3×10^{-4} SI (Uyeda *et al.* 1963; Hrouda 1982; Borradaile 1988; Archanjo *et al.* 1995). In general, ferromagnetic granites (Mt-bearing) have a specific susceptibility about three orders of magnitude higher than that of paramagnetic mineral species (Borradaile 1988). We attribute the high average bulk susceptibility

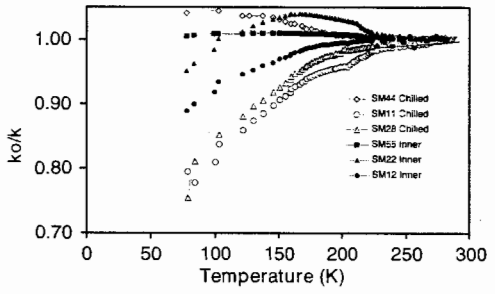


Fig. 8. Normalized reciprocal magnetic susceptibility (ko/k) as a function of temperature. Ideal ferromagnetic curve would be a straight line (no change in susceptibility with temperature), ideal paramagnetic curve is a line described by the Curie-Weiss law ($k_{para} = C/T - \theta$, where C = Curie constant, θ = paramagnetic Curie temperature; T = temperature in kelvin).

(13.46×10^{-3} SI $\pm 9.48 \times 10^{-3}$ SI) to magnetite as the principal mineral phase contributing to the overall susceptibility of the Stoddard Mountain intrusion (Table 1). AF demagnetization of the NRM reveals that MD magnetite grains of intermediate coercivity are characteristic of the chill zone rocks, as suggested by median destructive fields (MDF) of the NRM, when it is essentially single component in nature, between 10 and 20 mT (Fig. 9). The typical behaviour of the inner zone rocks implies a dominance of somewhat higher coercivity phases, with MDFs of about 30–80 mT (Fig. 9). Stepwise thermal demagnetization of duplicate specimens essentially give the same demagnetization results and yield laboratory unblocking temperatures between 540 °C and 635 °C (Fig. 5). This behaviour is indicative of the presence of low Ti to nearly pure magnetite and maghemite, both of which carry a important fraction of the natural remanence and low-coercivity magnetizations are likely viscous in origin (O'Reilly 1984; Dunlop & Özdemir 1997).

IRM acquisition curves show a narrow spectrum of responses, ranging from MD-magnetite-dominated behaviour to that influenced by somewhat higher-coercivity PSD magnetite and, on the basis of other observations, some maghemite (Fig. 9). All samples are likely dominated by a cubic phase (magnetite-type curves) and these show steep acquisition curves and nearly reach saturation by 0.20 T. Most specimens reach complete saturation by 1.25 T, yet some did not reach a stable plateau by 1.25 T (e.g. SM6, Fig. 9). Typically, IRM acquisition curves climb steeply below 0.1 T and display moderately to strong inflections around 0.15 T. The curves for the inner zone rocks reach saturation by about 0.15 T, and these rocks are likely dominated by a lower coercivity assemblage of grains than that typical of the chill zone rocks, which display curves that reach saturation at slightly higher fields than inner zone rocks. This result may indicate that multiple generations of ferromagnetic phases are present in the chilled zone rocks, which is consistent with the general multicomponent composition of the NRM (Fig. 5).

Back-field IRM acquisition curves show a very narrow range of responses, with coercivity of remanence values between about 0.030 T and 0.065 T. Chill zone rocks yield values between 0.045 T and 0.065 T, and inner zone rocks yield values between 0.030 T and 0.065 T (Fig. 9).

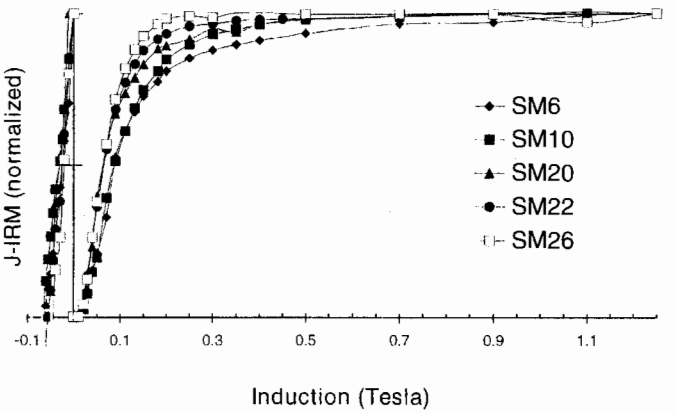


Fig. 9. Representative normalized IRM acquisition and back-field IRM demagnetization curves.

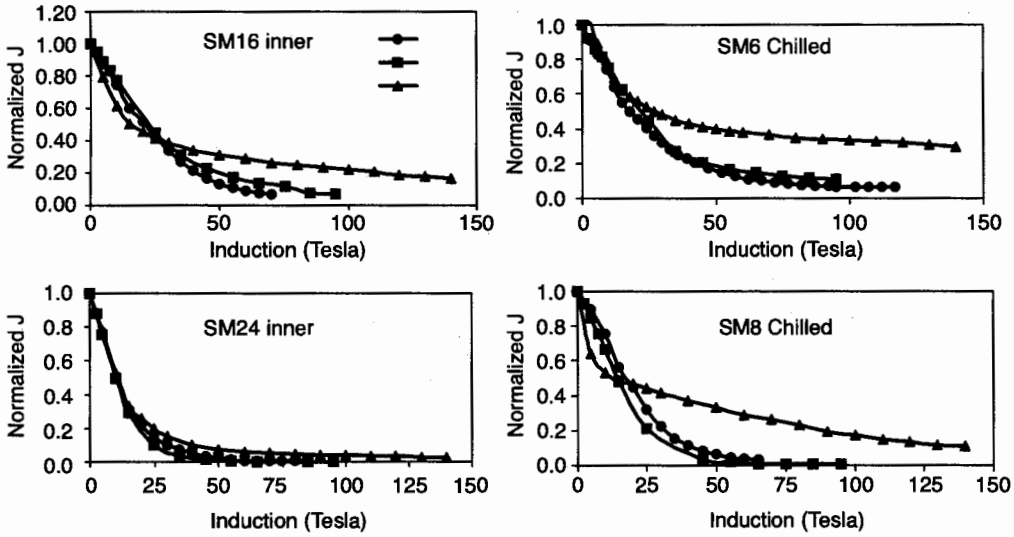


Fig. 10. Modified Lowrie-Fuller test (Johnson *et al.* 1975) comparing AF demagnetization response of natural remanent magnetization (NRM), anhysteretic remanent magnetization (ARM) and saturation isothermal remanent magnetization (SIRM).

Modified Lowrie–Fuller test

Following the modified Lowrie–Fuller test (Johnson *et al.* 1975) the AF decay of normalized NRM, ARM and SIRM was compared. The test is based on the experimental observation that normalized AF demagnetization curves of weak-field TRM (i.e. NRM) and strong-field TRM (i.e. SIRM) have different relationships for SD and MD grains of magnetite (Dunlop & Özdemir 1997). Laboratory investigations typically use weak-field ARM as a proxy for TRM, and we adhere to that procedure. In large MD grains, SIRM requires larger destructive fields than ARM to reach the same normalized level (i.e. the MDF of SIRM is larger than MDF of ARM). AF demagnetization of strong- and weak-field IRM show a limited spectrum of responses. All curves suggest MD magnetite-dominated behaviour, with a limited range in grain size likely responsible for carrying the remanence (Fig. 10). The chill zone rocks generally yield curves where ARM is more resistant than SIRM. In inner zone rocks, SIRM is generally more resistant to AF demagnetization than ARM.

Continuous susceptibility vs. temperature

Continuous susceptibility vs. temperature measurements on a Kappabridge KLY-3S instrument from room temperature to about 700 °C were made to further evaluate the magnetic mineralogy. The measurements were conducted in air, because maintaining an AR atmosphere with high-susceptibility igneous rocks during the experiment does not yield significantly different results. Continuous susceptibility vs. temperature measurements show a consistent decrease in room temperature susceptibility, of about 5 to about 20%, after heating to near 700 °C (Fig. 11a). Inferred Curie temperatures range from about 560 °C to about 635 °C. In some cases, the presence of a second ferromagnetic phase (maghemite) is indicated by a decrease in susceptibility over a temperature range of 300 °C to 400 °C in the heating curve, with no indication of its presence in the cooling curve. Inferred Curie temperature ranges are similar for inner and chill zone rocks.

Bulk susceptibility with temperature

Bulk susceptibility measured on a Sapphire Instruments SI-2 system between steps in the progressive thermal demagnetization of the NRM shows similar results for chilled and inner zone rocks, although some variations do occur (Fig. 11b). Most inner zone rocks are characterized by a rapid drop (about 40 to 60%) between 350 °C and 450 °C, suggesting that maghemite is converted to hematite (Fig. 11b) (Dunlop and Özdemir 1997). Magnetite is inferred to remain stable to about 580 °C (i.e.

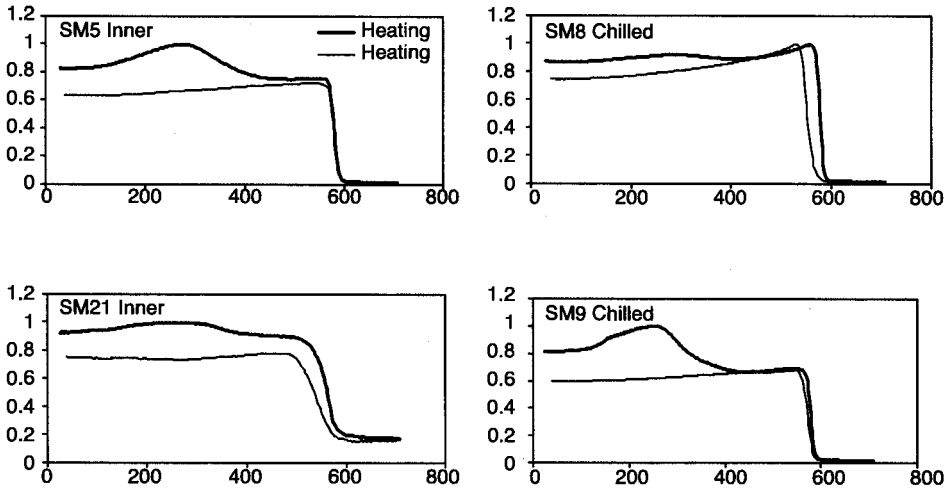


Fig. 11. Continuous susceptibility vs. temperature experiments and bulk susceptibility as a function of thermal demagnetization. (**a**) Continuous bulk susceptibility measurement during heating (solid lines) to about 700 °C and cooling (dotted lines) to 25 °C. (**b**) Normalized bulk susceptibility during stepwise progressive thermal demagnetization of the NRM.

<10% change in χ to 580°C). At higher temperatures (>580 °C), susceptibility decays uniformly by another 20%, suggesting that maghemite continues to break down and is an important carrier of the NRM of the rocks. Most chill zone rocks are characterized by a rapid drop (about 10%) between 350 °C and 450 °C, with magnetite remaining stable to about 580 ° C (i.e. <5% change in χ to 580 °C). No change in susceptibility occurs at temperatures above 580 °C. It is likely that maghemite is converted to hematite in the oxygen-rich environment of the furnace and that magnetite carries the natural remanence up to 580 °C, with most of the NRM unblocked by this temperature.

Thermal demagnetization of three component IRM

The test of Lowrie (1990) revealed considerable heterogeneity in the overall magnetic mineralogy of the materials studied from the Stoddard Mountain intrusion (Fig. 12). In some cases, the IRM acquired at 3.0 T is similar to (but never greater than) that acquired at 0.3 T (e.g. specimen 13F, Inner Zone, north part of pluton). The 3.0 T IRM persists to well above 600 °C, indicating that hematite is an important phase in these rocks, despite the fact that it does not carry appreciable remanence, based on thermal demagnetization. The 0.3 T IRM component varies in intensity by over an order of magnitude, and is always greater than that acquired at 0.03 T. For some specimens (e.g. 12G, Inner Zone, north part of pluton) this IRM component is not completely unblocked by about 580 °C, suggesting that fine-grained maghemite may be an important phase. Thermal demagnetization of specimens from samples of this site, for example, show that the NRM is fully unblocked by about 620 °C. Overall, we see no systematic relationship between inner zone and chilled zone rocks and their response to thermal demagnetization of three component IRM. For example, those sites from the chill zone that yield the south-directed and shallow inclination remanence (i.e. specimens 7D, 9F, 11B, 19E, and 26B) exhibit a range of behaviours comparable to those from sites in inner zone rocks.

Anisotropy of magnetic susceptibility

In rocks where the main magnetic minerals are MD low-titanium magnetites, as is the case for the Stoddard Mountain laccolith, the shape of the grains controls the AMS. The susceptibility anisotropy results from an alignment of elongated particles or a planar/linear distribution of the particles. All AMS measurements were performed prior to palaeomagnetic analysis, because AF treatment can impart a domain anisotropy superimposed on the shape anisotropy (Park *et al.* 1988; Potter & Stephenson 1990;

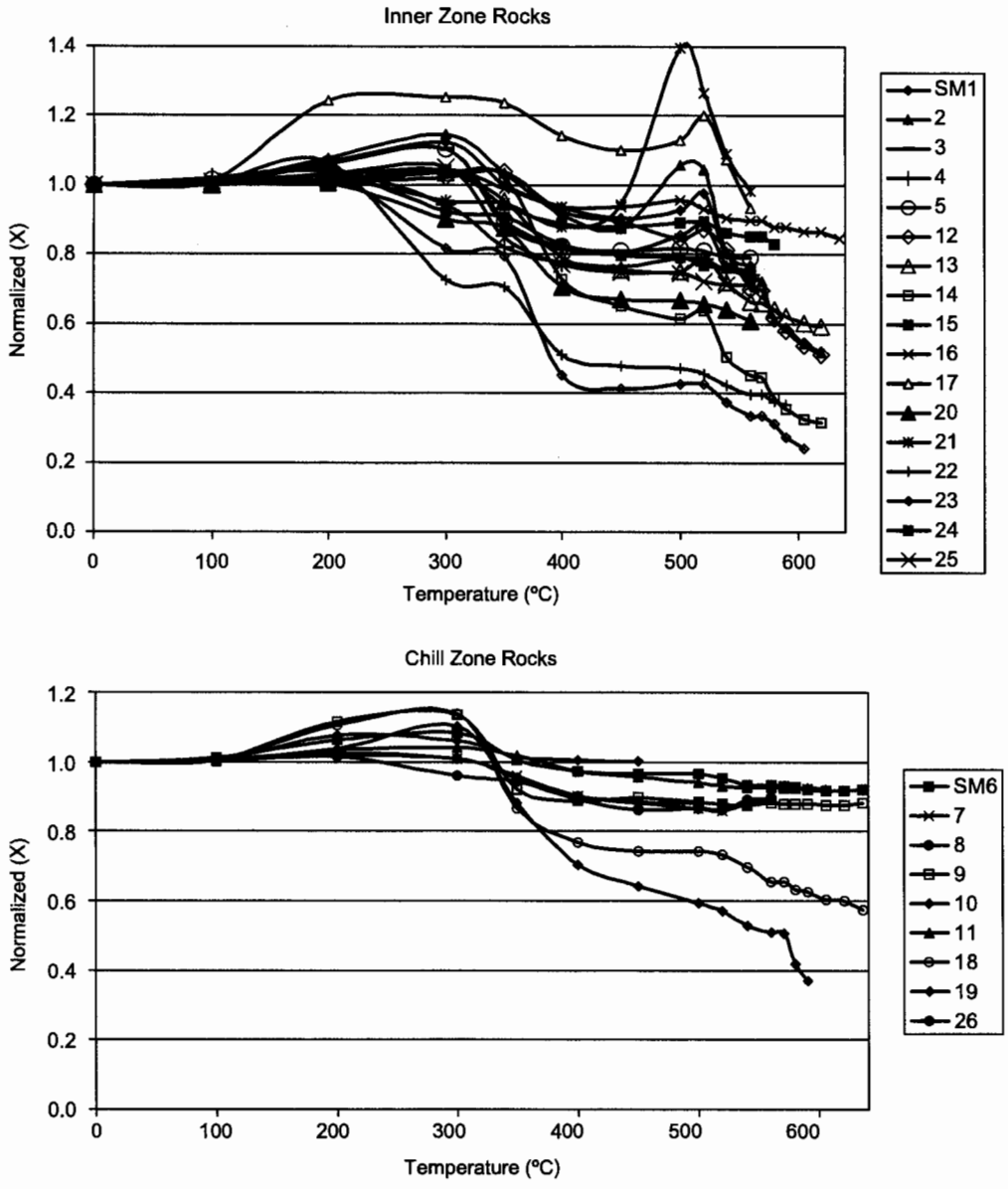


Fig. 11. Bulk susceptibility as a function of thermal demagnetization, and continuous susceptibility vs. temperature experiments.

Rochette *et al.* 1992), especially for low anisotropy rocks. In total, 959 specimens were analyzed from 76 stations with 763 specimens yielding interpretable results. Excluded specimens (196) were rejected due to a high within sample dispersion, meaning the individual specimen results from the same sample did not overlap at one standard angular deviation, or an individual specimen result was greater than 2σ from the overall mean of the station. In addition, 12 stations were rejected because of high between-specimen dispersion; the confidence interval about the three mean principal tensor axes were all greater than 15° (Fig. 13). The magnetic susceptibility (Km) magnitudes from 763 specimens vary from 31.41 to 1.21×10^{-3} SI,

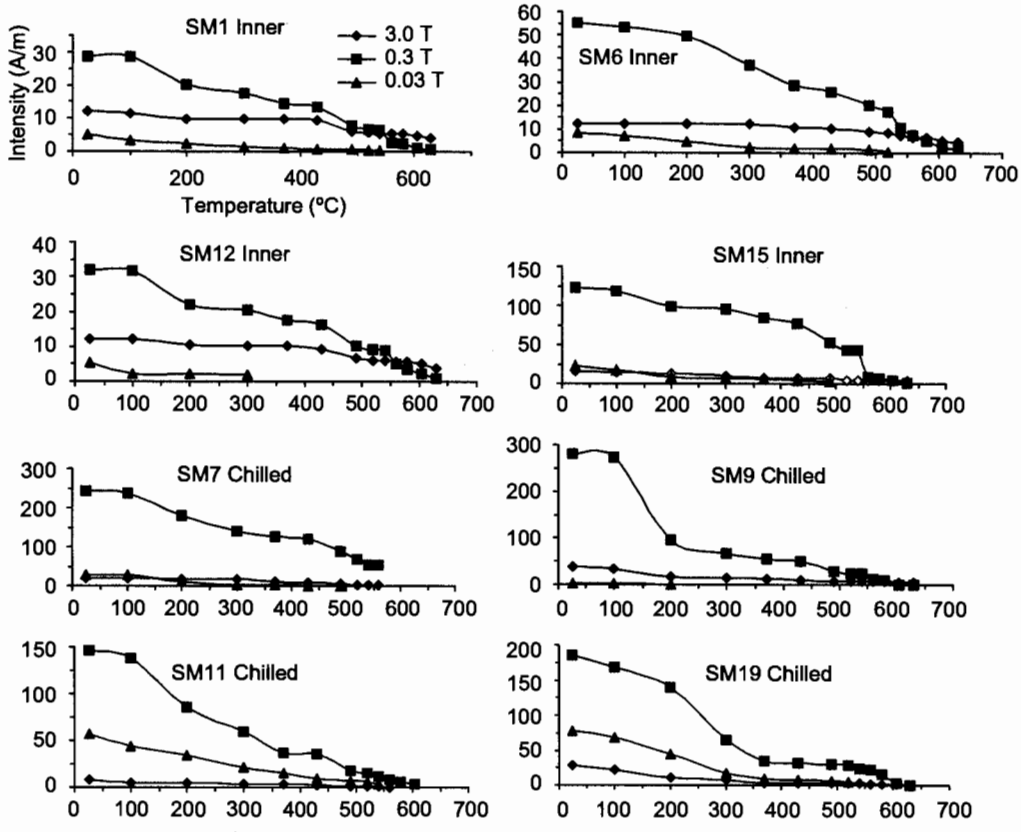


Fig. 12. Representative thermal demagnetization curves of three-component orthogonal IRM (Lowrie 1990).

magnetic anisotropy (Pj) ranges from 1.034 to 1.001, and magnetic ellipsoids are both prolate and oblate (Fig. 14a; data at the site level are summarized in Table 2). The bulk susceptibility of all samples is high and indicates that the magnetic susceptibility is mainly produced by ferromagnetic minerals (Rochette 1987; Bouchez 1997), with some contribution by Fe–Mg bearing silicates. The susceptibility magnitude is lower for inner zone rocks (mean of 11.02×10^{-3} SI) than that of chill zone rocks (mean of 21.18×10^{-3} SI). The anisotropy degree (Pj) is low, but also varied between inner and chill zone rock (mean inner = 1.014; mean chill zone = 1.018), and the average magnitude of Pj generally increases with the increase of the bulk susceptibility (Table 2). The shape of the AMS ellipsoid, defined by the parameter of T (Jelínek 1978), is not correlated with Pj (Table 2; Fig. 14b).

AMS fabrics, defined by magnetic lineations (K_1 axes) and magnetic foliation planes ($K_1 - K_2$ plane), are spatially variable through the general extent of the laccolith and are inferred to correlate with the magmatic fabrics in specific parts of the intrusion that developed during emplacement (Table 2; Fig. 15). In general, magnetic lineations throughout the inner zone have low to moderate plunge values with N–NE to S–SW trends, while the chilled zone localities are characterized by magnetic foliations or lineations that parallel the country rock contact (Fig. 15a). To simplify the description of the magnetic fabric distribution, the AMS data are divided into three zones, each of which reveals consistent lineations and/or planar fabrics: (1) chilled zone (19 stations), (2) northern zone (27 stations) and (3) southern zone (31 stations).

Chilled zone

In general, all chilled zone stations, regardless of their location on the margin of the intrusion, are characterized by a well-developed, steep to moderate dipping planar magnetic fabric that parallels the sub-circular contact with the

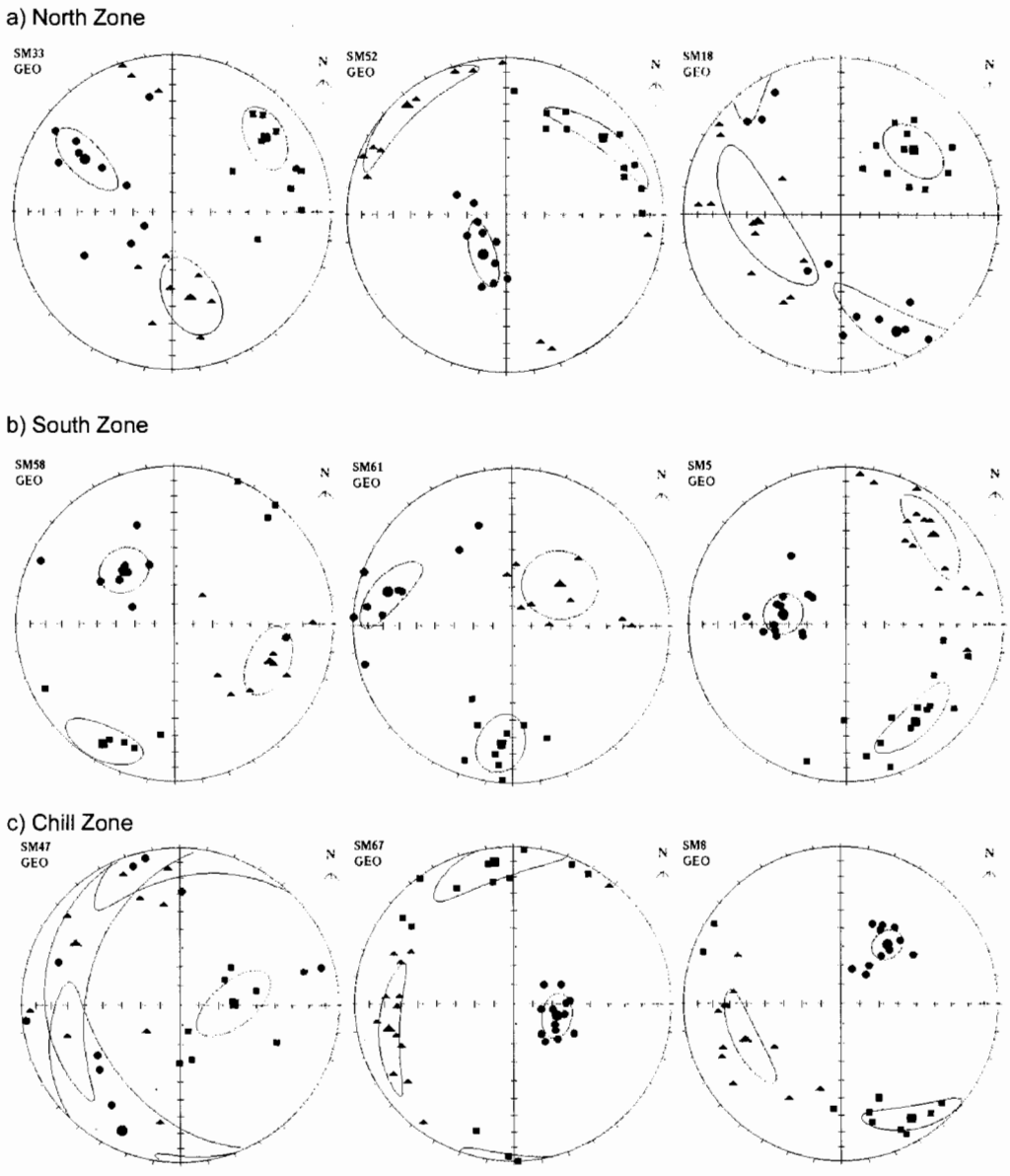


Fig. 13. Lower hemisphere equal area projections of AMS principal susceptibility axes (max K_1, intermediate K_2, and minimum K_3) for representative sites in the (**a**) chill zone, (**b**) northern zone, and (**c**) southern zone rocks of the Stoddard Mountain laccolith.

country rock (Fig. 15b), with magnetic lineations also trending roughly parallel to the country rock contact (Fig. 15b). One station was rejected due to a high between-sample dispersion. More specifically, magnetic lineations along the western margin of the laccolith trend roughly parallel to the contact with the country rock (NW–SE) and plunge gently (about 21°). A moderate- to steep-dipping planar magnetic fabric parallels the country rock contact at most stations (11/13 stations). Two stations near Richie Flat, however, yield magnetic lineations that trend east-west and have a moderate south-dipping planar fabric that are

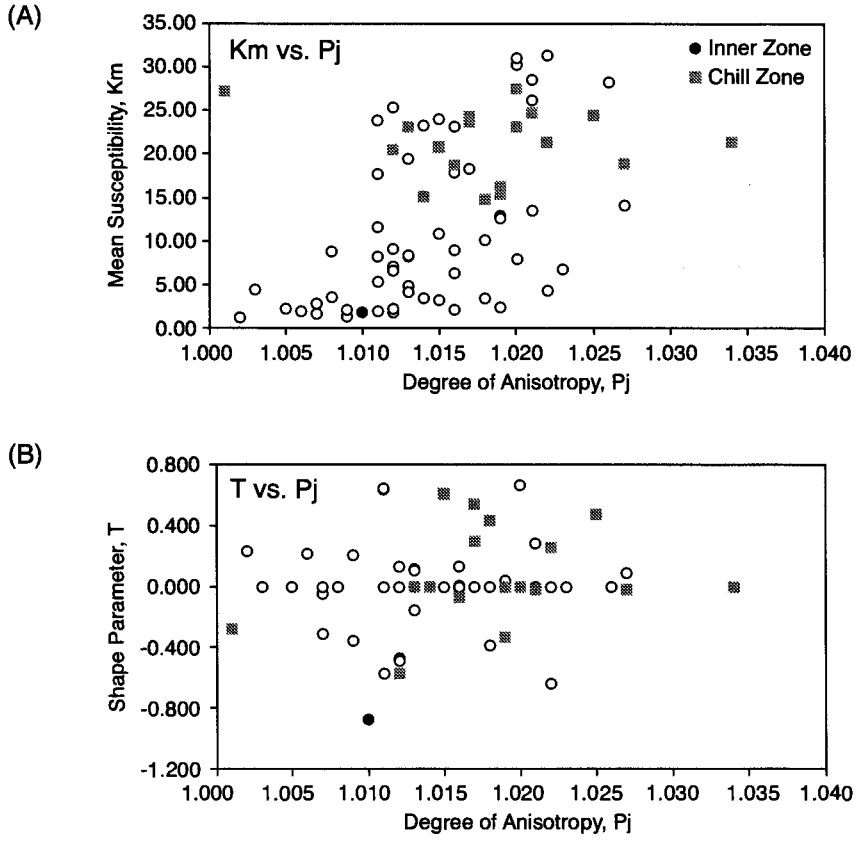


Fig. 14. Relationship between the degree of magnetic anisotropy, Pj, and (**a**) bulk susceptibility, Km (**b**) shape of the susceptibility ellipsoid, T.

perpendicular to the country rock contact (Fig. 15). The trend of the magnetic lineations near Richie Flat seem to reflect the flow pattern, while the remainder of the chilled zone stations seem to be influenced by the laccolith margin. Five stations on the NE and eastern margin of the intrusion yield a well-defined moderate to near vertical dipping planar fabric that parallels the country rock contact (Fig. 15b).

Northern zone

Of the 27 stations established in the northern zone, 20 stations yield interpretable results. Magnetic lineations in the northern zone trend generally E–NE and plunge gently to moderately (Fig. 15a), although two stations yield a SE trend and one station yields a NW trend with plunges for the three stations between 1° and 23° (Fig. 15a). Sub-horizontal planar fabrics, which are generally well defined, characterize stations along the traverse through the northern zone (Fig. 15b).

Southern zone

Of the 31 stations established in the northern zone, 27 stations yield interpretable results. Magnetic lineations in the southern zone trend south to SE and plunge gently to moderately. Four stations in the eastern-part of the laccolith yield magnetic lineations that trend nearly east towards the location from which the laccolith vented (Fig. 4; Fig. 15a). One station yields a magnetic lineation direction that trends NE and plunges gently (14°) and one station trends NW and plunges gently (15°). Overall, sub-horizontal planar fabrics characterize stations in the southern zone (Fig. 15b).

Discussion

Our study of the AMS and palaeomagnetism of the shallow level Stoddard Mountain intrusion shows that the petrographically distinct zones of the quartz monzonite exhibit different charac-

Table 2. Palaeomagnetic data from the Stoddard Mountain laccolith

Site	n/N	R	α_{95}	K	D	I	VGP LAT	LONG	RK Type
SM1	6\6	5.949	6.8	97.61	340.6	64.6	73.1	−165.3	Inner
SM2	7\7	6.952	5.4	125.73	340.7	60.9	74.5	179.3	Inner
SM3	5\7	4.956	8.1	89.85	348.9	65.7	76.8	212.1	Inner
SM4	5\6	4.966	7.1	116.45	357.1	52.2	84.7	94.1	Inner
SM5	6\6	5.948	6.8	97.05	0.9	54.8	87.6	47.8	Inner
SM6	6\6	5.932	7.8	73.95	179.3	−68.9	−75.1	64.9	Inner
SM7	6\6	5.758	15.1	20.70	181.2	21.1	−41.5	−114.9	Chilled
SM8	reject								Chilled
SM9	5\6	4.892	12.7	37.15	180.6	13.0	−45.8	−114.2	Chilled
SM10	6\6	5.918	8.7	60.88	182.7	12.8	−45.9	−117.3	Chilled
SM11	6\6	5.882	10.4	42.54	193.2	4.8	−48.2	−133.4	Chilled
SM12	6\6	5.904	9.4	51.97	336.6	46.9	68.3	137.8	Inner
SM13	4\6	3.990	5.4	287.87	344.5	48.4	74.7	124.2	Inner
SM14	5\6	4.966	7.1	116.36	350.3	51.1	80.2	115.3	Inner
SM15	6\6	5.973	4.9	118.20	349.2	55.4	81.2	140.1	Inner
SM16	6\6	5.851	11.7	33.56	349.6	51.0	79.7	125.3	Inner
SM17	5\6	4.971	6.5	139.79	10.7	49.5	78.6	10.3	Inner
SM18	7\7	6.924	6.9	78.69	147.1	−73.1	−60.0	32.4	Chilled
SM19	4\6	3.909	16.3	32.85	181.5	−12.3	−58.7	−116.3	Chilled
SM20	reject								Inner
SM21	6\6	5.952	6.6	104.26	349.3	55.7	81.3	151.2	Inner
SM22	5\6	4.957	8.0	93.06	358.1	55.2	87.6	108.2	Inner
SM23	6\6	5.969	5.3	159.46	354.8	58.8	85.5	184.4	Inner
SM24	5\6	4.980	5.5	197.95	349.1	55.9	81.2	152.9	Inner
SM25	6\6	5.987	3.5	376.10	355.0	57.8	86.0	171.0	Inner
SM26	6\7	5.909	9.1	55.06	172.4	−8.6	−56.1	−99.7	Chilled
SM28	5\6	4.9437	9.1	71.0093	187.5	11.4	−46.0	−128.5	Chilled
SM31	5\5	4.9778	5.7	179.9127	158.3	3.8	−45.6	−85.9	Chilled
SM44	5\5	4.8867	13.1	35.3153	174.7	−9.6	−56.8	−108.0	Chilled
SM45	7\7	6.886	8.4	52.6411	177.8	−30.9	−68.9	−111.9	Inner
SM66	4\5	3.9817	7.2	163.6567	174.1	3.4	−50.3	−108.6	Chilled
SM73	5\6	4.9591	7.8	97.6953	358.8	66.6	78.5	−121.7	Inner

Sites with (*) were rejected in this study. n/N = ratio of samples accepted for statistical analysis to samples collected; R = resultant vector length; α_{95} = 95% confidence interval about the estimated mean direction, assuming a circular Fisher distribution; κ = best estimate of (Fisher) precision parameter; D/I = in situ declination and inclination; VGP = in situ latitude and longitude of the virtual geomagnetic pole; RK Type, rock type, inner zone or chilled zone.

teristics in both their palaeomagnetic remanence behaviour and their anisotropy of magnetic susceptibility.

Palaeomagnetism

Palaeomagnetic data from the 30 sites yield two distinct populations of group means. Most chill zone rocks yield a south declination and shallow positive and negative inclination mean ($D = 178.5°$, $I = 0.9°$, $\alpha_{95} = 10.0°$, $k = 24.1$, $N = 11$). Most inner zone rocks yield a N–NW declination and moderate positive inclination mean ($D = 351.1°$, $I = 57.8°$, $\alpha_{95} = 3.7°$, $k = 82.2$, $N = 19$), which we interpret to be of normal polarity. Two of these sites yield a south-directed and moderate negative mean, essentially antipodal from the normal polarity results. The mean direction from inner zone rocks is statistically distinguishable from a Mid-Tertiary normal polarity expected direction ($D = 358°$, $I = 58°$; $D = 178°$, $I = -58°$) for west-central Nevada (based on the palaeomagnetic pole of Mankinen *et al.* 1987). The inner zone rocks may reflect a very small magnitude (<10°) anticlockwise rotation of the Stoddard Mountain laccolith, which is consistent with similar age rocks in the area (Hudson *et al.* 1998). Most sites in chill zone rocks, however, yield an unusual magnetization direction (south-directed and very shallow inclination) that is more than 90° from an expected Late Miocene normal polarity field and over 45° from an expected reverse polarity field. This discordant magnetization cannot be readily explained by

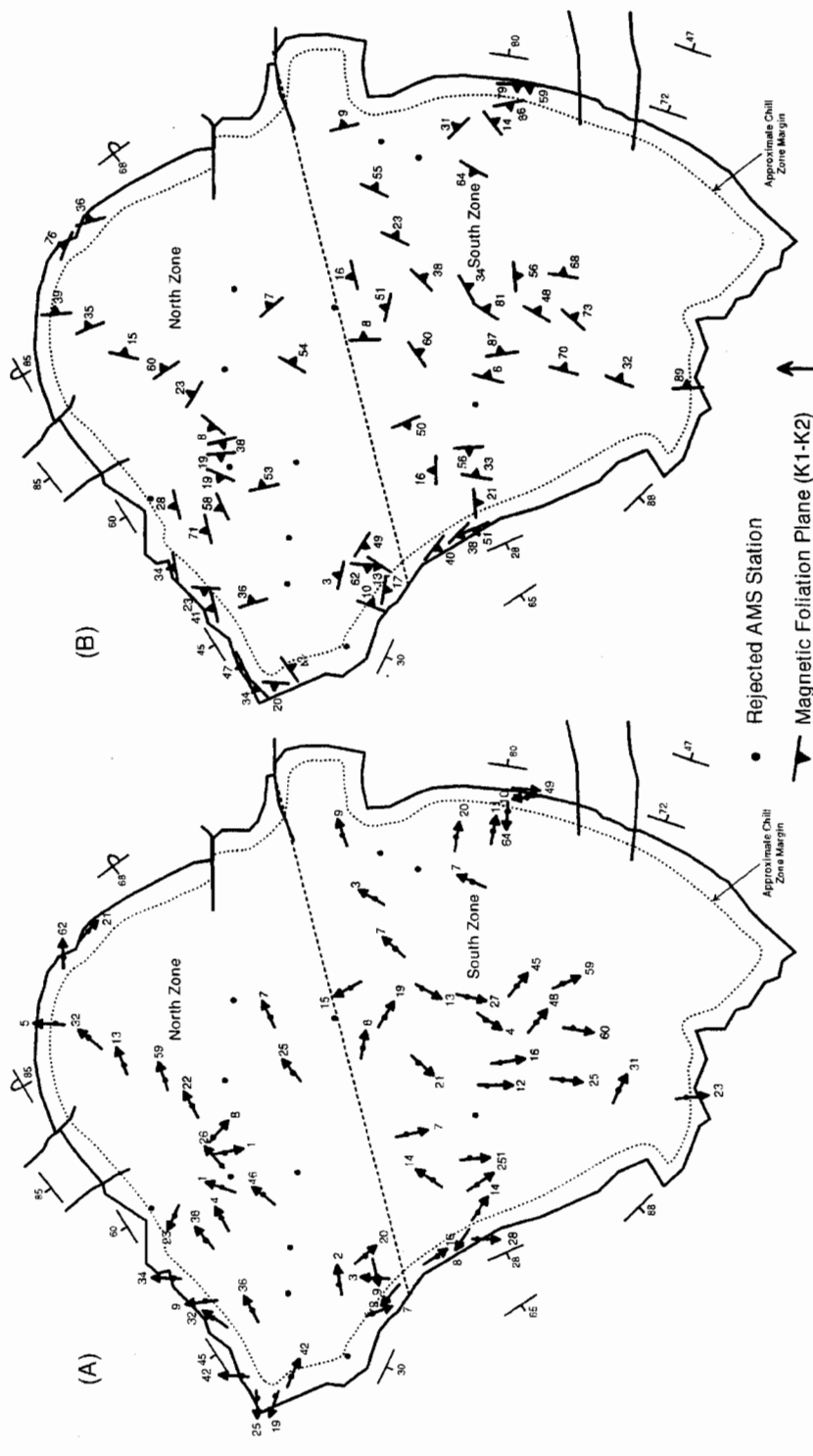


Fig. 15. Summary of accepted AMS data from the Stoddard Laccolith. (**a**) magnetic lineation (solid arrows), (**b**) magnetic foliation (solid line with triangle). Rejected stations are indicated by black circle. See text for rejection criteria.

some magnitude of tilting, or a complex combination of tilting and rotation, in particular because the data are from sites distributed around the intrusion. The near uniform directions from sites around the laccolith margin argue against a physical process, such as differential tilting, as a viable explanation of the discordant magnetization. Assuming that the magnetization characteristic of the chill zone rocks is a faithful record of the field, a plausible explanation is that these chill zone rocks record some form of short-lived transitional field direction. The rocks were emplaced at roughly 1 km depth and the chilled zone would have crystallized very rapidly relative to secular variation and lock-in a remanence direction quickly. If reasonable, then this explanation may imply that the chill zone rocks were emplaced early as a sill, which was later intruded by the main body of the intrusion, in a pre-magnetization acquisition state, as the laccolith inflated to its full extent. In this case, the chill zone rocks have been fractured by the main body of the intrusion and have been tilted, prior to magnetization acquisition, about a horizontal axis by a magnitude equal to the adjacent steeply dipping country rock. Subsequently, the chill zone rocks, which presumably would have cooled earlier than the inner zone rocks, acquired their unusual field direction. Additional investigation is needed to evaluate this possibility. Notably, inner zone rocks cooled over a sufficiently long period to time to capture both magnetic field polarities.

Anisotropy of magnetic susceptibility

The high magnetic susceptibility (average 11×10^{-3} SI) indicates that a ferromagnetic phase dominates the susceptibility in the Stoddard Mountain intrusion. In comparison, for example, Pyrenean granites yield susceptibilities two orders of magnitude less than the Stoddard Mountain intrusion (Gleizes *et al.* 1993). Because magnetite is the main carrier of the AMS, the fabric is defined by both the crystalline anisotropy and shape preferred orientation of these particles dispersed in the rock matrix (Stacey 1960; Uyeda *et al.* 1963). The directional regularity of the magnetic fabric, both within individual stations and between stations over several kilometres of the laccolith, suggests that an efficient mechanism operated to control the AMS. In reflected light studies, magnetite is slightly inequant in shape and associated with

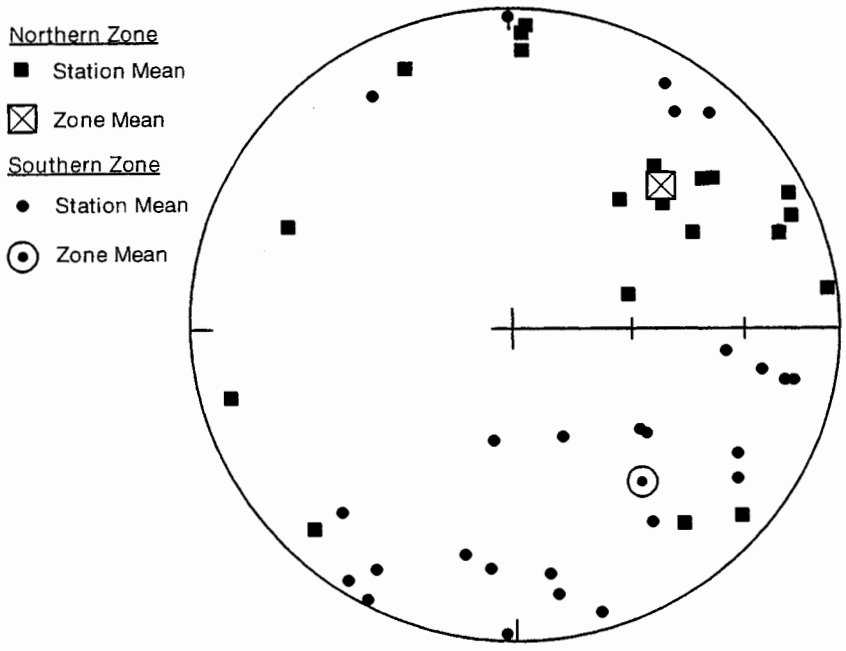


Fig. 16. Lower hemisphere equal area projections of maximum susceptibility axes (K_1) from stations in the north zone (black squares) and south zone (black circles). Solid symbols indicate station mean K_1 axes; square with X, mean K_1 direction of northern zone; open circle with dot, mean K_1 direction of southern zone.

biotite and amphibole, either as inclusions or along grain boundaries. The magnetite grains, when in contact with the biotite or amphibole, tend to have their long axes parallel to the crystallographic axes of these minerals (Fig. 3). It seems reasonable to propose that if biotite and amphibole grains have a shape preferred orientation (SPO) in the intrusion, then the alignment of the long axes of magnetite grains will control the magnetic fabric. This kind of 'mimetic' behaviour, where the orientation of magnetite is controlled by the SPO of biotite and amphiboles, has been recognized as an important mechanism for producing consistent AMS fabrics over large areas in other ferromagnetic intrusions (Hrouda *et al.* 1971; Archanjo *et al.* 1992; Launeau & Bouchez 1992; Cruden & Launeau 1994). It is likely that the AMS fabric in the Stoddard Mountain laccolith is controlled by ferromagnetic mineral phases in a preferred crystallographic orientation dictated by the paramagnetic mineral phases in the rock.

Inner zone

The AMS fabrics, which we infer to correlate with magmatic fabrics, are interpreted to reflect two principal flow paths (N–NE and roughly south) (Fig. 16). AMS lineations in the northern zone of the laccolith trend N–NE and plunge gently (mean flow direction = 044°/34°, $N = 20$) and in the southern zone lineations trend south and plunge gently (mean flow direction = 139°/30°, $N = 27$).

Chill zone

The AMS fabrics reflect a principal flow path defined by a well-developed foliation parallel to margin of the intrusion, except in the NW region, where the foliation is perpendicular to the margin of the intrusion and lineations trend east-west. We hypothesize that the significantly different AMS fabric in the chill zone, relative to the inner zone reflects an early, pre-inflation flow pattern that was controlled by the margin of the intrusion.

Emplacement mechanisms

Studies of several Late Oligocene to Early Miocene Colorado Plateau laccoliths suggest that magma emplacement occurred from below by a central axial feeder pipe or dyke(s) from which they grew both laterally and vertically (Corry 1988; Jackson & Pollard 1988; Ross 1998). In contrast, geological relations in the area surrounding the Stoddard Mountain laccolith suggest that this intrusion may have been fed laterally from near the Pinto Peak area west of Stoddard Mountain. We hypothesize that the magma flowed eastward beneath Richie Flat and around a southern barrier of Tertiary volcanic rocks and then spread laterally south to north and inflated into a laccolith (Fig. 18). In this model, inferred magma flow paths expected for the development of a laterally fed laccolith whose source area is to the west are considerably different from those for an axial fed laccolith (Fig. 17). In the axial feeder system (Fig. 17a), flow paths in cross-section vary from vertical near the central source to sub-horizontal near the base. In map view, a symmetric, near radial flow path would be expected as the pluton spread laterally in all directions as a sill. In contrast, for the lateral feeder system (Fig. 17b), magma flow paths would tend to be only slightly or moderately inclined given the high length to thickness ratio of the laccolith. In map view, the magma flow path would be asymmetric, tending to radiate outwards from a point source in dominantly one direction (i.e. eastward in the case of Stoddard Mountain laccolith). In map view, the expected flow directions are markedly different for the western side of the laccolith. The magnetic anisotropy axes and thus magnetic fabrics determined here are interpreted to represent indicators of the principal axes of strain undergone by the Stoddard magma at the end of its emplacement. The fabric that is preserved likely represents the latest stage of magma emplacement prior to solidification. In map view, the orientations of AMS axes in the Stoddard intrusion are spatially variable across the short, east-west axis of the intrusion (Fig. 18). In the north part of the laccolith, AMS lineations trend north to NE (mean flow direction trend/plunge of about 044°/34°, $N = 20$) and in the south part of the laccolith, lineations trend S–SE (trend/plunge of about 139°/30°, $N = 27$). The dominance of NE, east, and SE trending directions in the north, east, and south regions, respectively, is best explained by the lateral intrusion model (Fig. 17b). The plunge of K_1 is generally sub-horizontal to moderate. The main body of the laccolith lacks any vertically oriented magnetic lineations, which we infer to indicate the absence of a source area feeding the pluton from below. In the chill zone regions of the pluton margin, magnetic foliations are everywhere parallel to the contact except along the inferred source area in the NW part of the intrusion (Fig. 15).

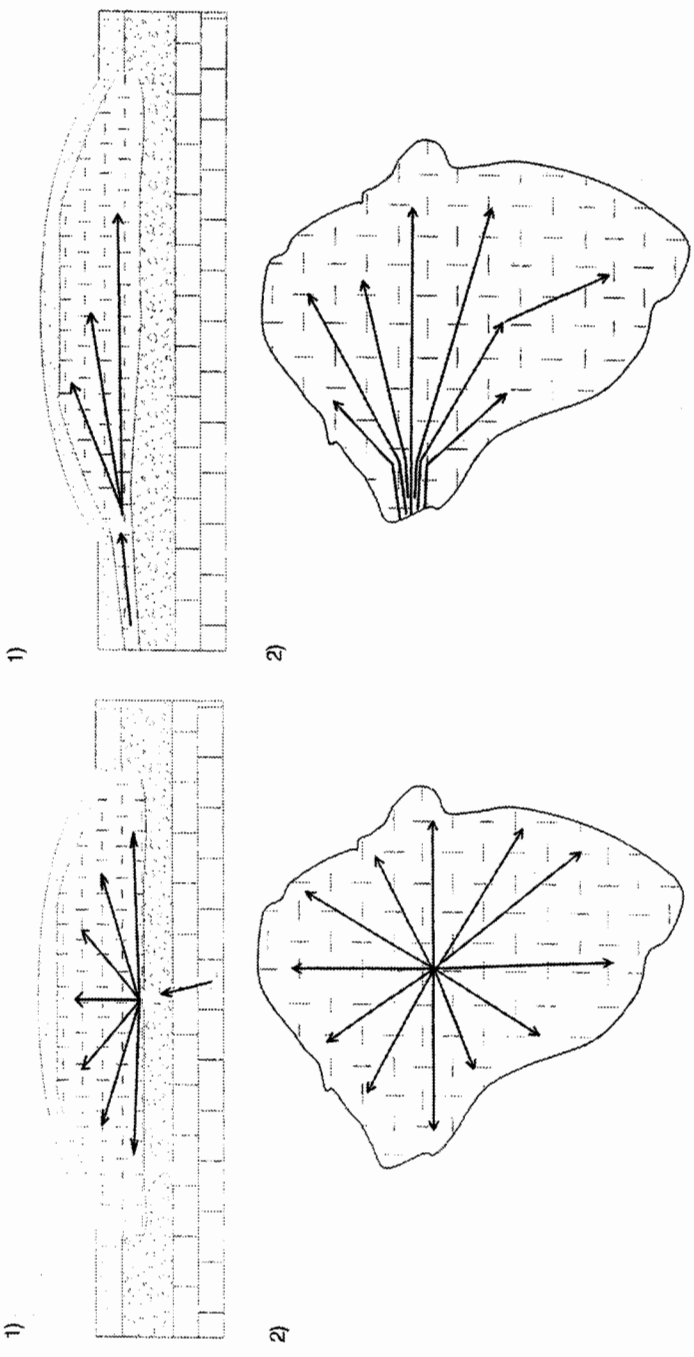


Fig. 17. Idealized schematic models for laccolith formation. (**a**) Magma flow paths for a laccolith fed by an axial feeder dyke system; (**b**) Magma flow paths for a laccolith fed by a lateral feeder system. Arrows delineate expected magma flow paths in (1) cross-section and (2) map view.

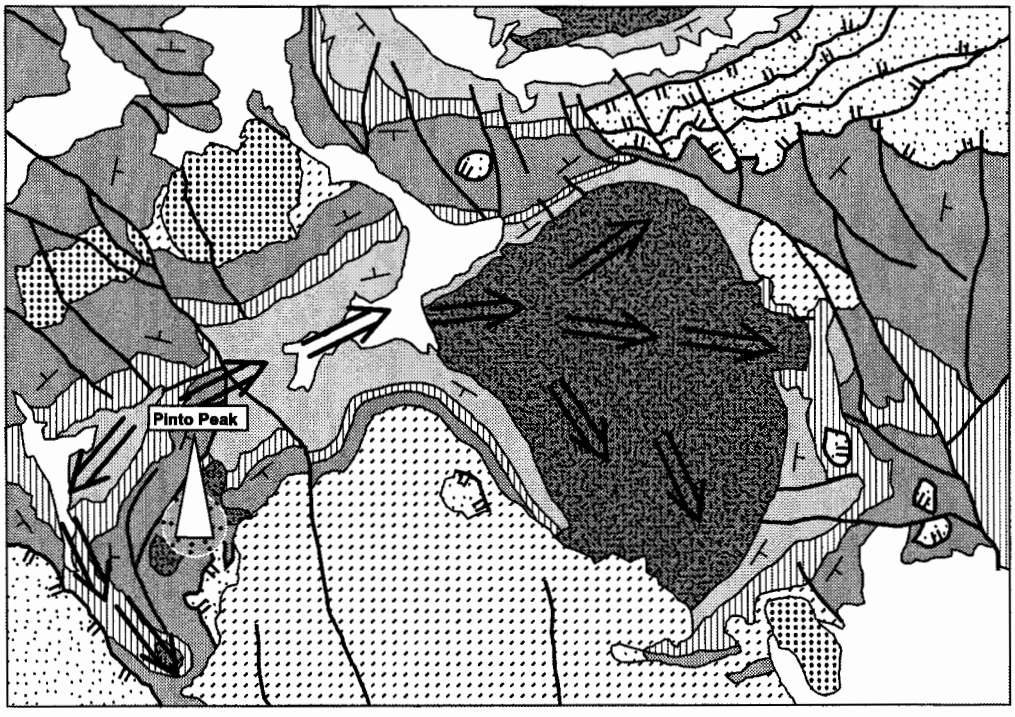


Fig. 18. Interpretative diagram depicting our inferred magma flow model that led to the formation of the Stoddard Mountain laccolith, based on the available AMS data and geological field relations. Arrows indicate hypothesized magma flow direction.

Here, the magnetic foliation is approximately east-west, perpendicular to the contact and likely parallel to the magma flow direction. The observed variation in trend of magnetic fabrics and the overall shallow plunge of magnetic lineations is most consistent with the lateral feeder hypothesis for the emplacement and growth of the Stoddard Mountain laccolith (Figs 17b, 18).

Conclusions

Based on palaeomagnetic and AMS data reported here we reach the following conclusions bearing on the emplacement history of the Stoddard Mountain laccolith. The palaeomagnetic data from the inner zone rocks reveal that the main mass of the intrusion has not been affected by any appreciable deformation, either local tilting or vertical axis rotation, since emplacement. Based on field relations and thin section observations, we suggest that the magmatic flow fabrics are predominately controlled by a preferred orientation of paramagnetic mineral phases. Superimposed on the paramagnetic fabric is one attributable to fine-grained, MD to PSD magnetite grains, which control the AMS directions, contribute substantially to the overall high magnetic susceptibility (greater than 10^{-3}SI), and carry the remanent magnetization. The Stoddard Mountain laccolith displays well-organized magnetic fabrics that reflect magma flow during emplacement (Fig. 18). Our rock magnetic data are interpreted to indicate the dominance of low-titanium coarse-to intermediate grained magnetite as the high-susceptibility fraction within the intrusion. MD magnetite grains control the low-field magnetic fabric of the intrusion. The MD nature of magnetite, with its tendency to statistically align its magnetic moments parallel to its grain elongation suggests a correlation between the shape of the grains and the magmatic fabric. The similarity of the magnetic and magmatic fabrics is a consequence of the fact that magnetite is likely mimicking the orientation of the ferromagnesian phases in the rock.

The magnetic anisotropy data correspond well with predicted magma flow directions associated with a laterally fed shallowly-emplaced laccolith.

The overall absence of NW and SW-trending, and steeply plunging magnetic lineations argues against emplacement via a central axial feeder system. Further AMS studies in the Iron Axis laccoliths may prove valuable for resolving conflicting data on magma emplacement models (i.e. Three Peaks area: Van Kooten 1988) and for better understanding the relationship between laccolith emplacement and ore genesis.

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